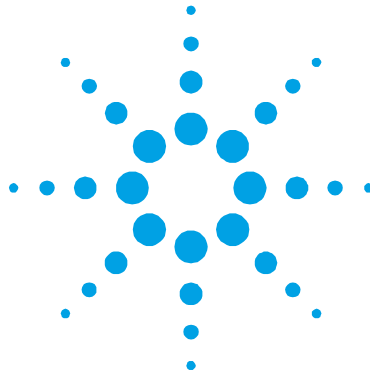




TwisTorr 74FS AG Rack Controller

Models

X3508-64301, X3508-64022

**Manuale di istruzioni
Bedienungshandbuch
Notice de mode d'emploi
Manual de instrucciones
Manual de instruções
Bedrijfshandleiding
Instruktionsbog
Bruksanvisning**

**Instruksjon manual
Ohjekäsikirja
Felhasználói kézikönyv
Podrecznik instrukcji
Návod k použití
Návod na obsluhu
Priročnik za navodila
User Manual**

87-901-055-01

07/2018



Agilent Technologies

Notices

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Manual Part Number

Publication Number: 87-901-055-01

Edition

Edition 07/2018

Printed in ITALY

Agilent Technologies Italia S.p.A.

Vacuum Products Division

Via F.lli Varian, 54

10040 Leini (TO)

ITALY

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CAUTION

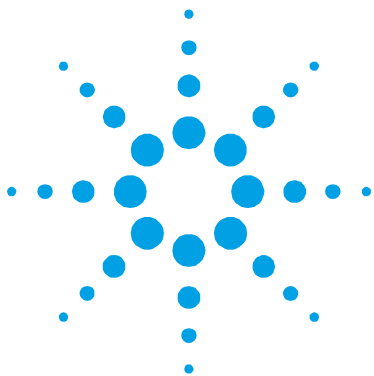
A **CAUTION** notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a **CAUTION** notice until the indicated conditions are fully understood and met.

WARNING

A **WARNING** notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in personal injury or death. Do not proceed beyond a **WARNING** notice until the indicated conditions are fully understood and met.

TwisTorr 74 FS AG Rack Controller





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Original Instructions



General Information

This equipment is destined for use by professionals. The user should read this instruction manual and any other additional information supplied by Agilent before operating the equipment. Agilent will not be held responsible for any events occurring due to non-compliance, even partial, with these instructions, improper use by untrained persons, non-authorised interference with the equipment or any action contrary to that provided for by specific national standards.

The TwisTorr 74 FS AG Rack controllers are micro-processor-controlled, solid-state, frequency converter with self-diagnostic and self-protection features.

Controller features:

- Front / Remote / Serial Operation
- 24Vdc pump fan cooling drive
- Vent valve drive
- Pump speed reading after stop command (stop speed reading)
- Pressure reading
- Input voltage auto setting.
- Profibus interface (optional)

The following paragraphs contain all the information necessary to guarantee the safety of the operator when using the equipment. Detailed information is supplied in the appendix "Technical Information".

This manual uses the following standard protocol:

WARNING!



The warning messages are for attracting the attention of the operator to a particular procedure or practice which, if not followed correctly, could lead to serious injury.

CAUTION!

The caution messages are displayed before procedures which, if not followed, could cause damage to the equipment.

NOTE

The notes contain important information taken from the text.

Storage

When transporting and storing the controllers, the following environmental requirements should be satisfied:

- temperature: from -20 °C to + 70 °C
- relative humidity: 0 – 95 % (without condensation)

Preparation for Installation

The controller is supplied in a special protective packing. If this shows signs of damage which may have occurred during transport, contact your local sales office. When unpacking the controller ensure that it is not dropped or subjected to any form of impact. Do not dispose of the packing materials in an unauthorized manner. The material is 100 % recyclable and complies with EEC Directive 85/399.

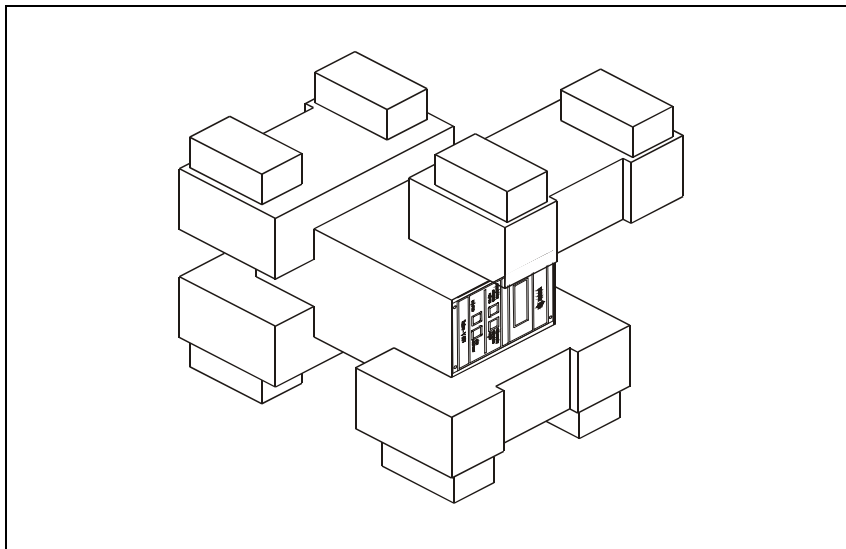


Figure 1 Controller Packaging

Installation

WARNING!



The TwisTorr controller is designed for indoor use only and must be powered with 3-wire power cord (see orderable parts table) and plug (internationally approved) for user safety. Use this power cord and plug in conjunction with a properly grounded power socket to avoid electrical shock and to satisfy CE requirements. High voltage developed in the controller can cause severe injury or death. Before servicing the unit, disconnect the input power cable.

NOTE

The TwisTorr controller can be used as a bench unit or a rack module, but it must be positioned so that free air can flow through the holes. Do not install or use the controller in an environment exposed to atmospheric agents (rain, snow, ice), dust, aggressive gases, or in explosive environments or those with a high fire risk.

During operation, the following environmental conditions must be respected:

- temperature: from +5 °C to +45 °C;
- relative humidity: 0 – 95 % (without condensation).

See the appendix “Technical Information” for detailed Information about the above mentioned and the other connections, and about the options installation.

Use

This paragraph describes the fundamental operating procedures. Detailed information and operating procedures that involve optional connection or option are supplied in the paragraph “USE” of the appendix “Technical Information”.

Make all vacuum manifold and electrical connections and refer to TwisTorr pump instruction manual before operating the TwisTorr controller.

WARNING!

Refer to the relevant pump manual to operate it in a safe way before starting.



NOTE

The input signal J1 connector should be left in position including the shipping links if no external connections are made. The forepump and TwisTorr pump can be switched on at the same time.

Controls, Indicators and Connectors

The following paragraph illustrates the Controller control panel and interconnection panel. More details are contained in the appendix “Technical Information”.

Front Panel Description

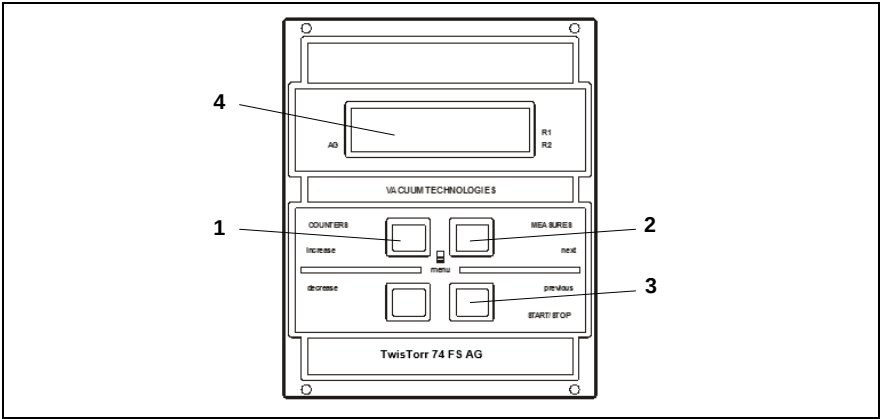


Figure 2 Controller TwisTorr 74 FS AG Front Panel

1	Keyboard push-button to recall on the display the cycle number, cycle time and pump life.
2	Keyboard push-button to recall on the display the “pump current”, “pump temperature”, “pump power” and “rotational speed”. It is always active regardless of the operating mode selected. Push-buttons 1 and 2, if pressed together for at least 2 seconds, put the controller in a routine where it is possible to program some operation parameters.
3	Keyboard push-button for START, STOP/RESET mode selection. It is active only when the front panel operation has been selected. By pressing once the starting phase begins; if pressed again it stops the pump. If the pump has been stopped automatically by a fault, this push-button must be pressed once to reset the controller and a second time to restart the pump.
4	LCD back-lighted alphanumeric display: dot matrix 2 lines x 16 characters.

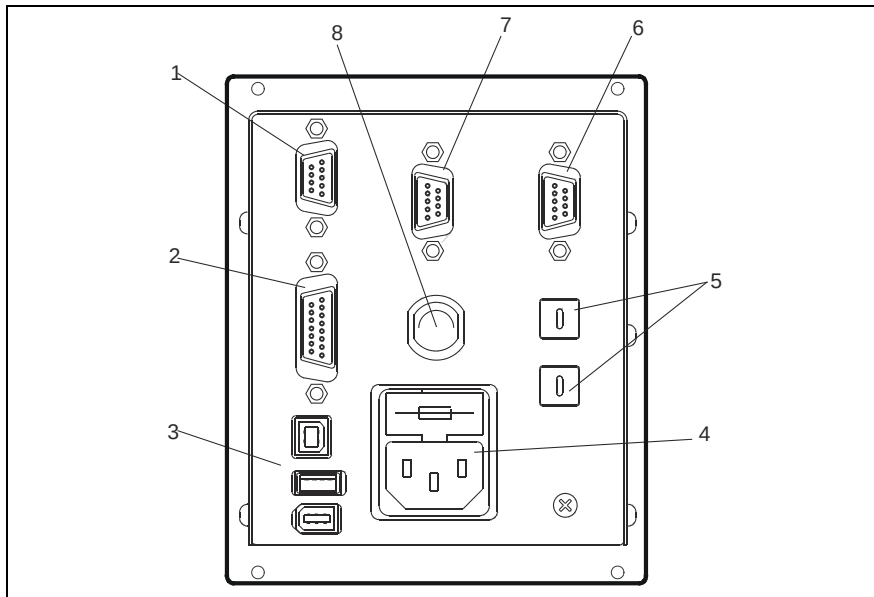


Figure 3 Controller TwisTorr 74 FS AG Rear panel

1	Logic input signals connectors (mating connector supplied with link).
2	Logic output signals connector and pump frequency monitor plus programmable analog output.
3	Power output connector (24 Vdc for pump cooling fan, Vent Valve and Gauge).
4	Controller power entry module consisting of mains fuses, mains socket and EMC filter.
5	Rotary switches for Profibus address setting (Optional).
6	Profibus (Optional).
7	Remote control serial connector.
8	Pump cable (5m long).

Use Procedure

Controller Startup

To startup the controller plug the power cable into a suitable power source.

Starting the Pump

To start the pump, confirm the interlock input (shorting pins 3÷8 on P1) and press the START pushbutton on the controller front panel.

Pump Shutdown

To shutdown the pump press the STOP push-button on the controller front panel.

Maintenance

The TwisTorr 74 FS AG Rack series controller does not require any maintenance. Any work performed on the controller must be carried out by authorized personnel.

When a fault has occurred it is possible to use the Agilent repair service. Replacement controllers are available on an advance exchange basis through Agilent.

WARNING!

Before carrying out any work on the controller, disconnect it from the supply.

If a controller needs to be scrapped, it must be disposed off in accordance with the specific national standards.

Disposal

Meaning of the "WEEE" logo found in labels

The following symbol is applied in accordance with the EC WEEE (Waste Electrical and Electronic Equipment) Directive. This symbol (**valid only in countries of the European Community**) indicates that the product it applies to must NOT be disposed of together with ordinary domestic or industrial waste but must be sent to a differentiated waste collection system. The end user is therefore invited to contact the supplier of the device, whether the Parent Company or a retailer, to initiate the collection and disposal process after checking the contractual terms and conditions of sale.



For more information refer to:

<http://www.agilent.com/environment/product/index.shtml>

Error Messages

The controller will self-diagnose some error causes and will display the following messages:

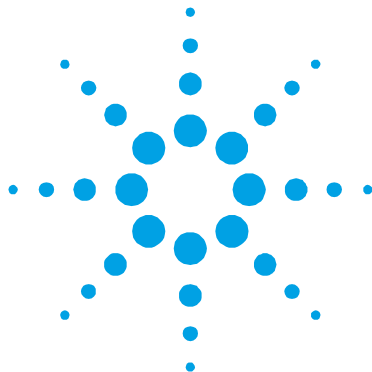
Tab. 1

MESSAGE	DESCRIPTION	REPAIR ACTION
CHECK CONNECTION TO PUMP	Wrong connection between the pump and the controller.	Check connection between controller and pump.
	OR The pump temperature is below 0°C.	Press the START push-button twice to start the pump.
WAITING INTERLOCK	The interlock signal of P1 connector is activated by an interruption of the link between pin 3 and 8 of J1 connector, or because the external interlock signal is open.	Connect pin 3 to pin 8 of P1 connector, or close the external interlock signal.
PUMP OVERTEMP.	The upper bearing/pump temperature exceeds 60°C.	Wait until the temperature decreases below threshold value. Press the START push-button twice to start the pump.
CONTROLLER OVERTEMP.	The controller environment temperature exceeds 70°C.	Wait until the temperature decreases below threshold value.
	OR The controller radiator temperature is above 60°C.	Press the START push-button twice to start the pump.

16 Instructions for Use

Error Messages

MESSAGE	DESCRIPTION	REPAIR ACTION
TOO HIGH LOAD	The current drawn by the pump is too high.	Check that the pump rotor is free to rotate. Press the START push-button twice to start the pump.
SHORT CIRCUIT	A short circuit is present on the pump connector.	Check connections and short circuits between pump and controller. Press the START push-button twice to start the pump.
SYSTEM OVERRIDE	The pump is stopped by an emergency stop signal provided via a remote contact.	Remove the controller power cable and check the emergency condition. Then reconnect the power cable and press the START push-button twice to start the pump.
OVERVOLTAGE	Controller power supply circuitry is faulty or the Controller received an unexpected excessive voltage.	Press the START push-button twice to start the pump. Should the message still be present, call the Agilent service.
POWER FAIL	Failure in the power supply section.	Contact Agilent for Maintenance.



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TwisTorr 74 FS AG Rack Controller Description

Two controller models are available (see orderable parts table):

- with RS 232-485 (X3508-64301)
- with RS & Profibus (X3508-64022)

The controller is a solid-state frequency converter with the following features:

- Drives the pumps of the 74 FS AG family.
- Powers the pump cooling fan
- Drives the vent valve.
- Provides and acquires the pressure of the wide range gauge.
- Remote I/Os compatible with the previous version.
- Speed reading after stop command.

See the following diagram for the connections and orderable parts.

17 Technical Information

TwisTorr 74 FS AG Rack Controller Description

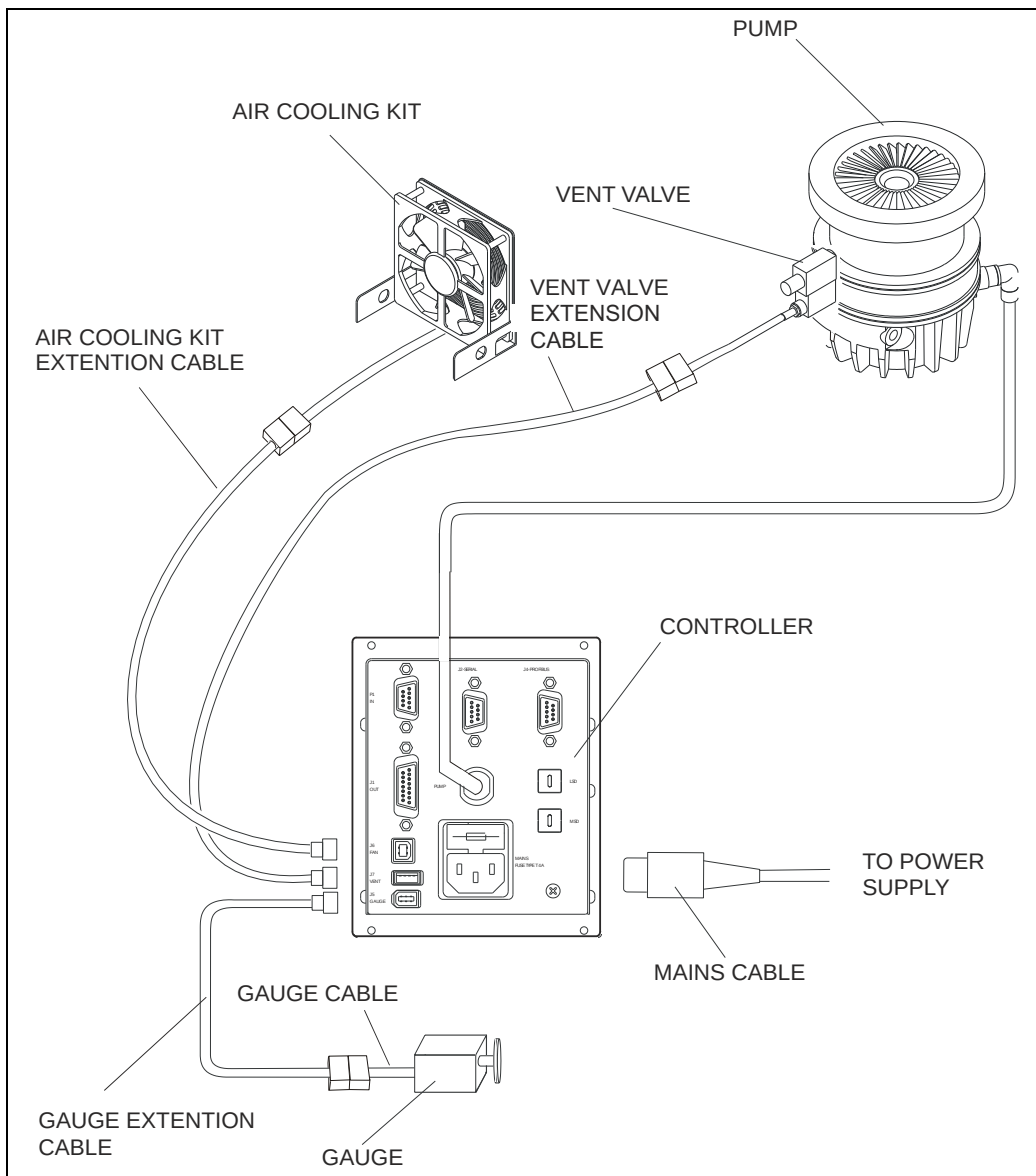


Figure 4 TwisTorr 74 FS AG Rack Controller – Connections

Controller Specifications

Tab. 2

Input:	
Voltage	100 to 240 Vac (voltage fluctuation +/- 10%)
Frequency	50 to 60 Hz
Power	450 VA
Fuse	
2 x T4 A (slow blow) 250 V	
Output:	
Voltage	75 Vac
Frequency	1167 Hz
Power	100 W (Ramp), 70 W (Normal and Autotuning)
Environment storage	
-20 °C to +70 °C 0 – 95 %	
Operating	
+5 to +45 °C 0 – 90 %	
Applied norr.	
EN 61010 -1 EN 55011 (Class-A) EN 61000-3-2, EN 61000-3-3, EN 61000-6-2, EN 61000-4-2,3,4,5,6,11, 8	
Weight:	
1.7 Kg (3.2 lbs)	
Pollution Degree	
2	
Installation Category	
II	
Acoustic noise (ISO7779)	
LpA < 70 dB(A), normal operation, operator position	

NOTE

The controller supplied voltage must be easily disconnected (through a circuit bracker or having access to the power supply cable plug).

Controller Outline

The outline dimensions for the TwisTorr 74 FS AG Rack Controllers are shown in the following figure.

NOTE

The TwisTorr controller can be used as a bench unit or as a rack module, in any case it must be positioned so that free air can flow through the housing holes.

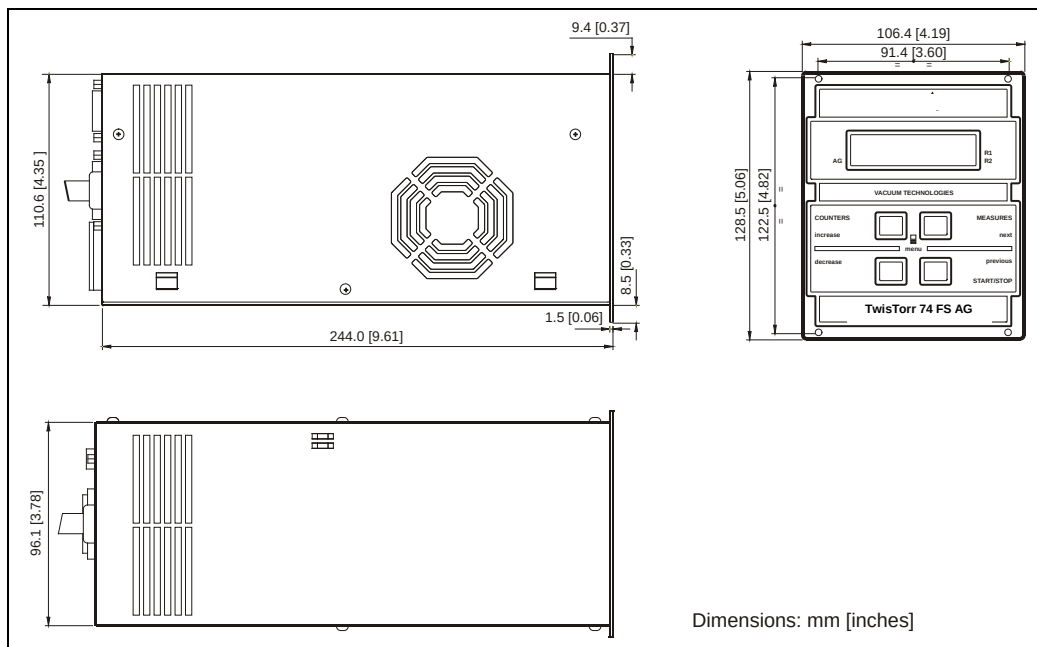


Figure 5 Controller outline

Controller Connection

J6 – Fan

The 24 Vdc, 150 mA maximum output current will be supplied according to the selected working mode (see win 143 on the serial protocol description paragraph). The default working mode is external fan active except in case of override error status.

This connection has to be used with TwisTorr 74 FS pump cooling fan (see orderable parts table).

Adapter cables are available to connect other fans.

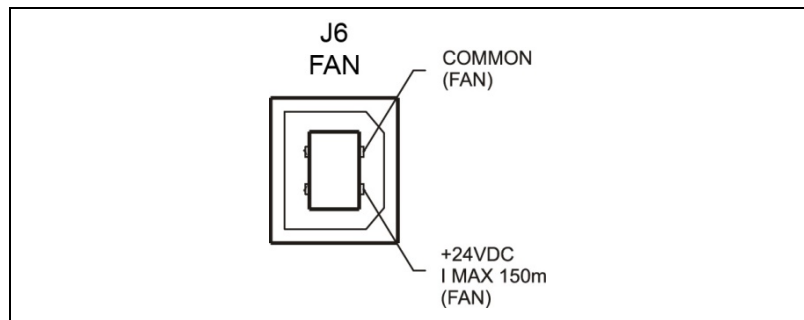


Figure 6 Fan Connector

J7 – Vent

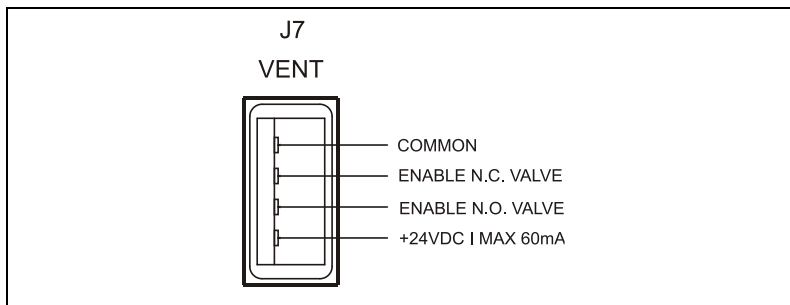


Figure 7 Vent Connector

This is the connection to control the optional vent valve (see orderable parts table). The vent valve can be driven by controller automatically or by serial line.

If you have other vent valves installed (110 Vac powered), it must be replaced (see orderable parts table).

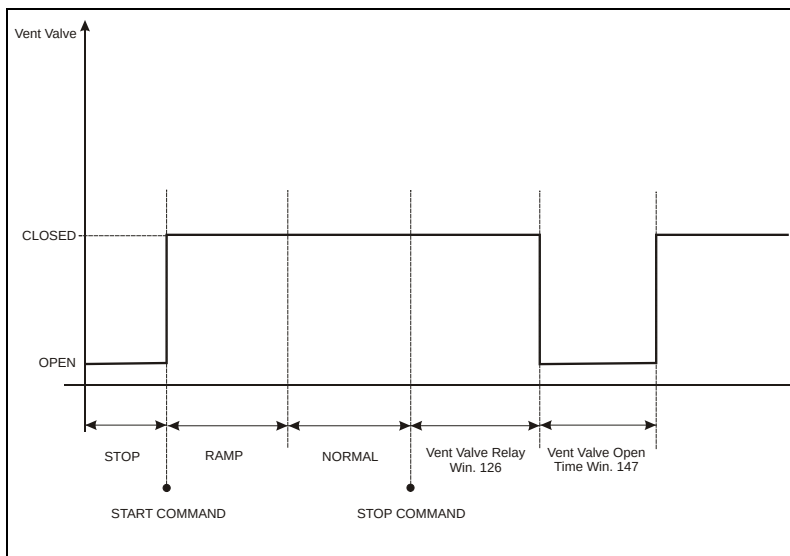


Figure 8 Vent Valve diagram in “Auto” mode

P1 – Input

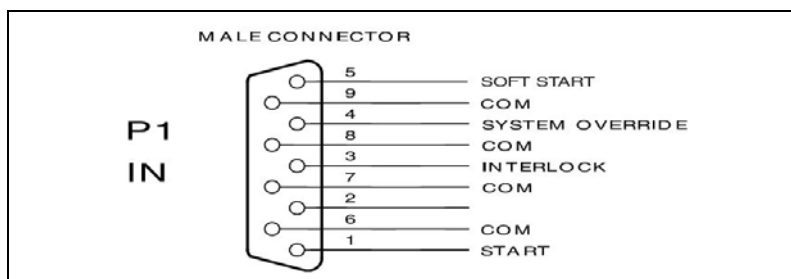


Figure 9 P1 input connector

All the logic input to the controller have to be connected to P1 mating connector. The provided P1 mating connector is equipped with pin 3 and pin 8 shorted to enable the pump start. The connections should be done with AWG 24, (0.24 mm²) or smaller wire. The following table describes the signals available on the connector.

Tab. 3

Pins	Name	Description	Electrical	Note
1-com	Start/Stop	Contact closed = Start Contact open = Stop	Optically isolated Vlow, max = 10v I _{max} = 5mA I _{min} (on) = 2mA	Active only in Remote Mode operation
3-com	Interlock	Contact closed = Active Contact open = Inactive If active allows the pump running, if inactive stops the pump.	Optically isolated Vlow, max = 10v I _{max} = 5mA I _{min} (on) = 2mA	Active in all modes
4-com	System Override	Contact closed Contact open If the contact is open, it allows the pump running; if the contact is closed, it stops the pump and all the others connected devices (Fan, Vent).	Optically isolated Vlow, max = 10v I _{max} = 5mA I _{min} (on) = 2mA	Active in all modes
5-com	Soft Start	Contact closed= Active Contact open= Inactive If active before the start command activates the soft start procedure. To be used only at first pump run-up.	Optically isolated Vlow, max = 10v I _{max} = 5mA I _{min} (on) = 2mA	Active only in Remote Mode operation, and in STOP status.

17 Technical Information

Controller Connection

The following figure shows a typical contact logic input connection and the related simplified circuit of the controller.

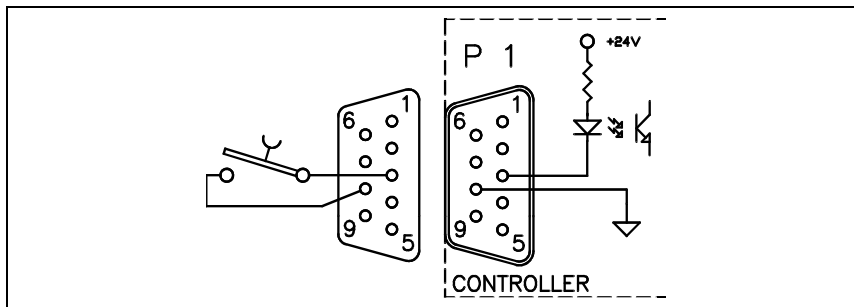


Figure 10 Typical logic input connection

J1 – Output

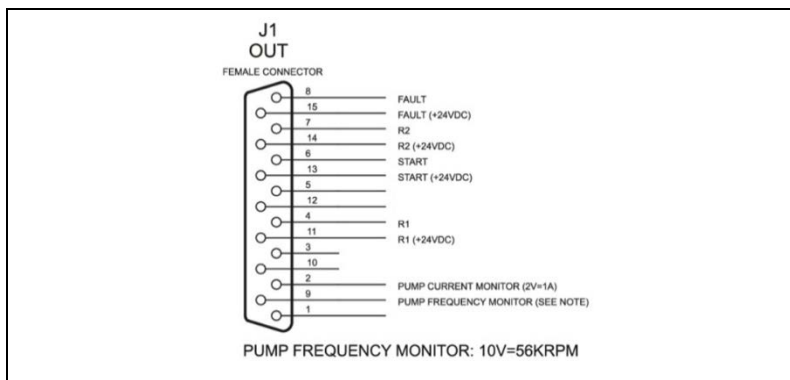


Figure 11 Logic output connector

All the logic output from the controller must be connected at J1 mating connector. The table 4 describes the signals available on the connector.

The following figure shows a typical logic output connection of a relay coil (other devices may be connected in the same way, e.g. a LED, a computer, etc.) and the related simplified circuit of the controller.

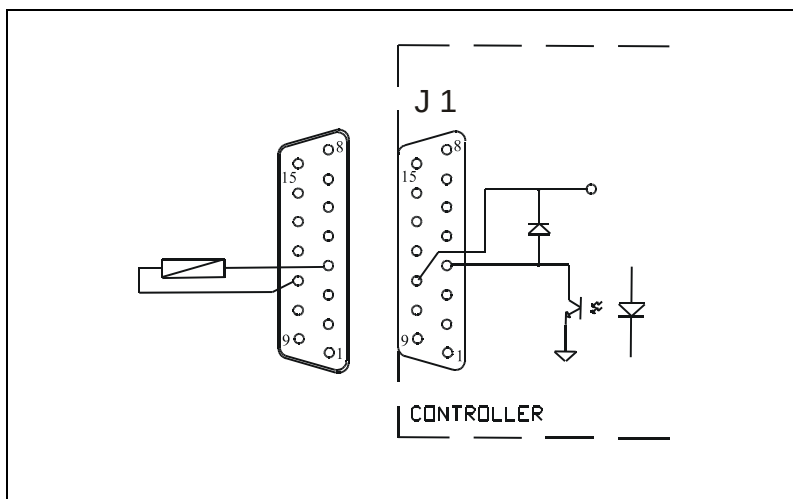


Figure 12 Digital Output Connection Example

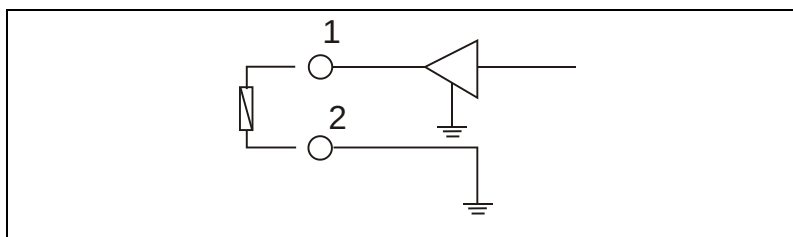


Figure 13 Analog Output Connection Example

17 Technical Information

Controller Connection

Tab. 4

Pins	Name	Description	Electrical	Note
15-8	Fault	0V = no fail 24V = fail	Optically isolated Vmax (=open) = 24V Vlow, max = 1,5V Imax = 60mA Pin-8 negative Pin-15 positive	See note 1. Connect a 10kohm resistor between the 2 pins before measuring voltage A STOP command (by front Panel or serial or Start/stop input) reset the FAIL status.
13-6	Start	Configurable output. Default setting: 24V pump in RAMP state, 0V = other state. See win 177 for further description.	Optically isolated Vmax (=open) = 24V Vlow, max = 1,5V Imax = 60mA Pin-6 negative Pin-13 positive	See note 1.
11-4	R1	Programmable Set Point Out. See note 2.	Optically isolated Vmax (=open) = 24V Vlow, max = 1,5V Imax = 60mA Pin-5 negative Pin-12 positive	See note 1.
14-7	R2	Programmable Set Point Out. See note 2.	Optically isolated Vmax (=open) = 24V Vlow, max = 1,5V Imax = 60mA Pin-7 negative Pin-14 positive	See note 1.

Pins	Name	Description	Electrical	Note
2-1	Programmable Analog Output	Programmable analog output. Can be related to: frequency, power, pump temperature (linear) or pressure read by the external gauge. The setting can be done by: front panel or serial line (win.111). 0V = 0Hz, 0W, 0°C 10V = 1167Hz, 70W, 100°C See following chart for more details	Optically isolated Rload, min = 10kohm Ripple Vpp,max = 200mV Precision = 5% Resolution = 40mV Pin-1 negative Pin-2 positive	
9-1	Frequency Analog Output	Analog output proportional to frequency (linear). 0V = 0Hz 10V = 1167Hz	Optically isolated Rload, min = 10 kohm Ripple Vpp,max = 200mV Precision = 1% Resolution = 40mV Pin-1 negative Pin-9 positive	

When the analog output is set to provide pressure signal, the voltage level is proportional to pressure level and it is generated by control unit elaborating the independent input by the active gauge.

NOTE 1

A load should be present to measure the voltage on open collector output; if not present connect 10 kohm resistor between the related pins.

NOTE 2

The R1 and R2 output behaviours can be configured through the serial communication as described in the serial communication window meaning table. The related windows are:

Tab. 5

	R1	R2
Set Point type (0÷4)	Win 101	Win 171
Set Point threshold value (expressed in Hz, W or s)	Win 102	Win 172
Set Point delay: time between the pump start and the set point check (seconds)	Win 103	Win 173
Set Point signal activation type: the signal can be "high level active" or "low level active"	Win 104	Win 174
Set point hysteresis (in % of value)	Win 105	Win 175
Set Point Pressure	Win 162	Win 176
Threshold Valid if min. 101 = 4		
Format X.X EsXX		
Where X = 0 to 9 s = + or -		

R1 – R2 Output Diagrams

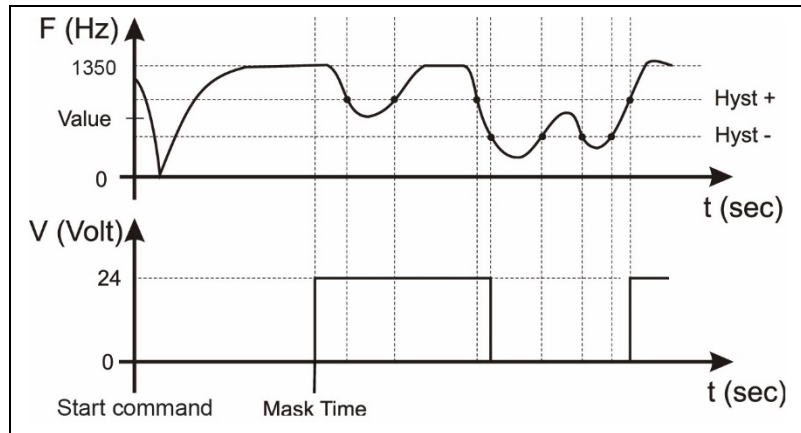


Figure 14 Diagram 1 – R1 or R2 related to frequency

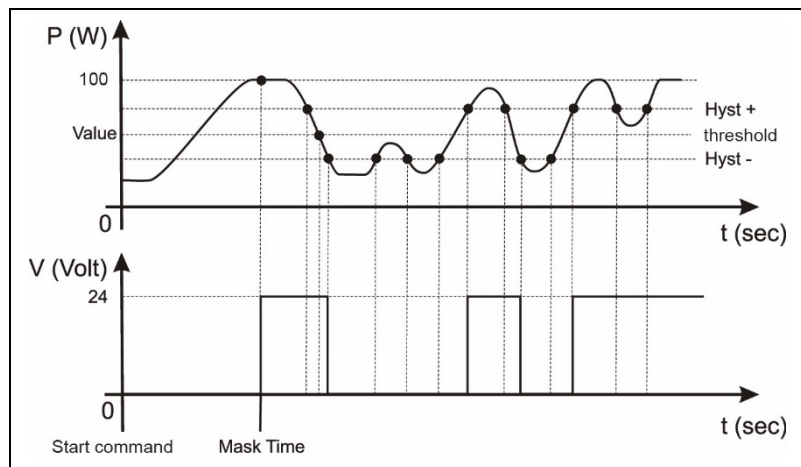


Figure 15 Diagram 2 – R1 or R2 related to power

17 Technical Information

R1 – R2 Output Diagrams

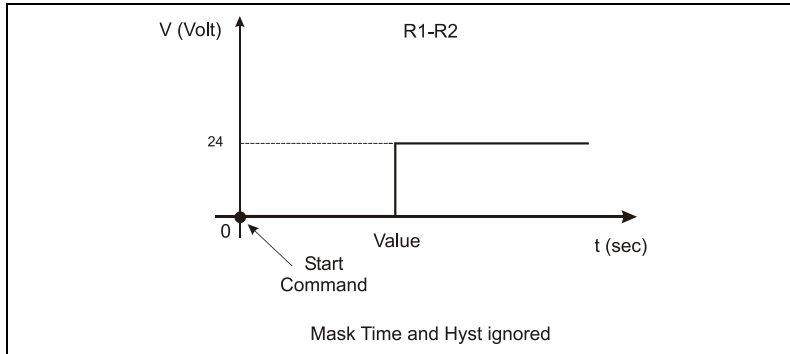


Figure 16 Diagram 3 – R1 or R2 related to time

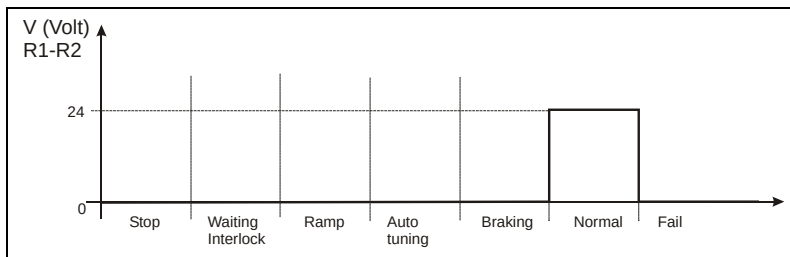


Figure 17 Diagram 4 – R1 or R2 related to Normal status

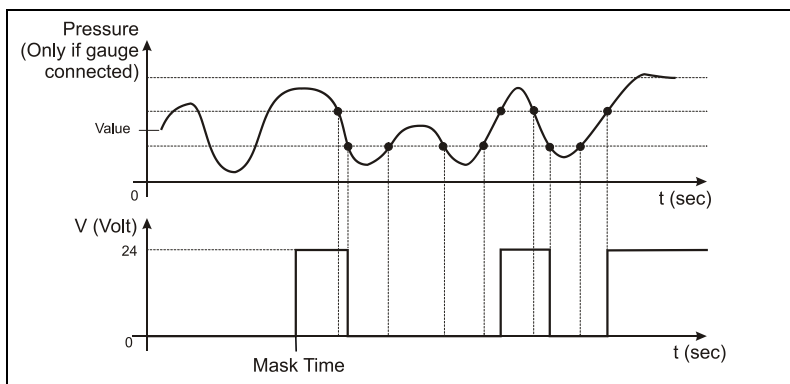


Figure 18 Diagram 5 – R1 or R2 related to pressure reading (if gauge connected)

Programmable Analog Output Diagrams

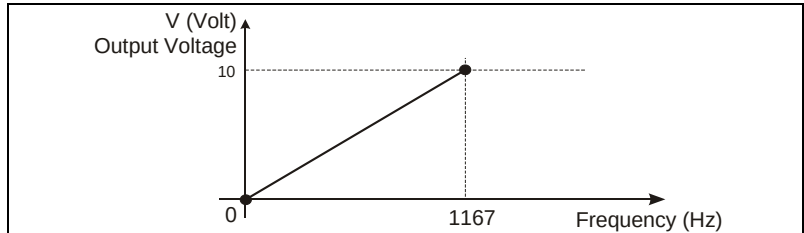


Figure 19 Diagram 1 – Programmable analog output related to frequency

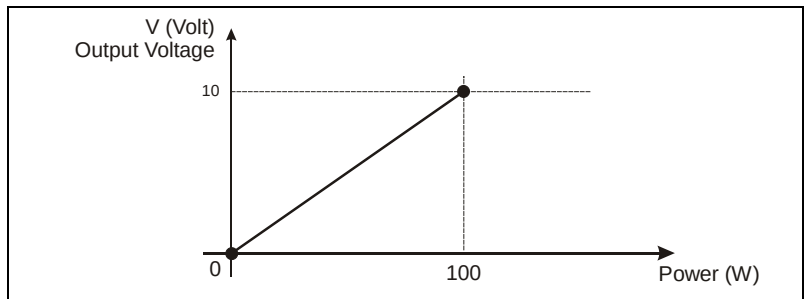


Figure 20 Diagram 2 – Programmable analog output related to power

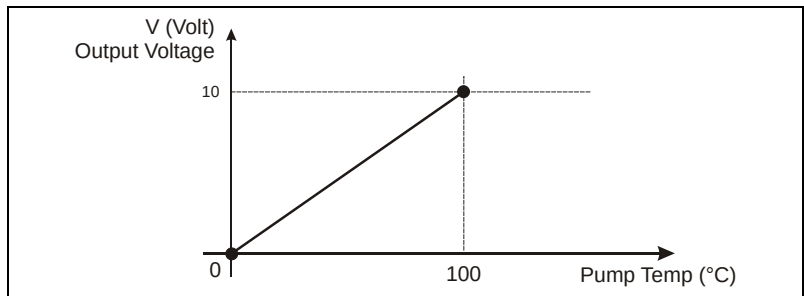


Figure 21 Diagram 3 – Programmable analog output related to pump temperature

Pump Connector

The controller is equipped with a 5 meter long cable. The following figure shows the controller output connector configuration where pins:

- **A-F** = pump temperature sensor
- **B-C-D** = 76 V AC 3-phase output to pump motor stator
- **E** = EARTH PROTECTION

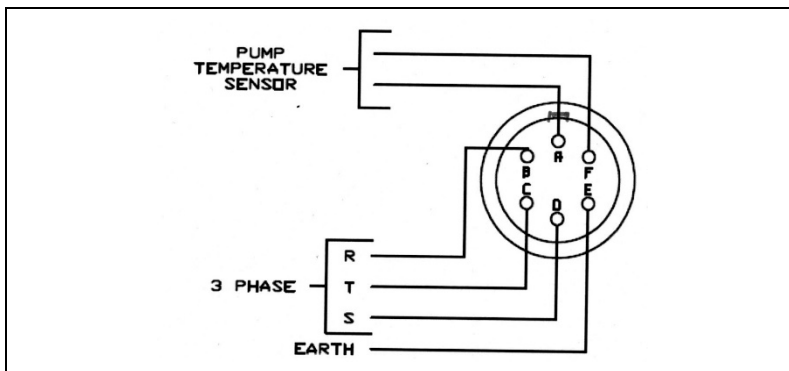


Figure 22 Controller-to-Pump Connector

J5 – Gauge Connector

This connector is available to connect the Agilent active gauges. The controller recognizes automatically the gauge connected by reading the signal “identification” in the connector. Following table shows all supported gauges:

Gauge model	Name showed in the display
FRG 700/702	FRGa
IMG 500	IMG
PVG 500/502	PVGa
IM 500	IM
CDG 500 T1000	CDG3
CDG 500 T0100	CDG2
CDG 500 T0010	CDG1
CDG 500 T0001	CDG0

The controller gives the error “gauge error” if the gauge connected is not supported. The pressure can be read via display, serial line (win. 224) or REMOTE I/O on Programmable Analog Out.

If the Programmable Analog Output is configured to show the pressure read by an external gauge, the controller doesn’t replicate exactly the signal coming from the gauge; it converts the gauge signal in the range from 0 to 10V.

The conversion formula is the following:

$$V_{out} = \frac{V_g - V_{g_min}}{V_{g_max} - V_{g_min}} \times 10$$

Where:

V_g is the gauge output voltage;

V_{out} is the Analog Output voltage;

$V_{g_min} \div V_{g_max}$ is the gauge voltage measuring range (these values can be drawn from the gauge technical specifications).

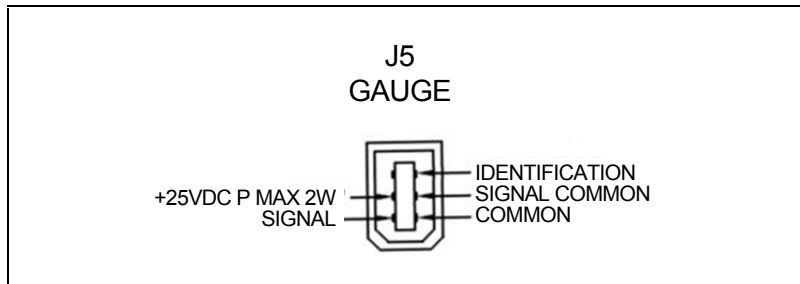


Figure 23 Gauge Connector

With the serial communication you can use the following windows:

Win 163	Pressure measurement unit (0=mbar, 1= Pascal, 2=Torr)
Win 224	Pressure reading with the format X.X E XX
Win 257	Gauge status (read only): 0 → when a gauge is not connected 1 → when a gauge is connected 2 → when the pressure read is under range or when the gauge give an internal error 3 → when the pressure read is under range or when the gauge give an internal error 4 → when the “identification” is present but not known by the controller
Win 267	Gauge power (read/write): If you need to turn off the gauge at certain condition you can use this window 0 → gauge off 1 → gauge on (default) 2 → gauge normally off. It turns on when the set point 1 is activated. If the set point 1 type (win 101) is selected as “pressure” the gauge is always on. 3 → gauge normally off. It turns on when the set point 1 is activated. If the set point 2 type (win 171) is selected as “pressure” the gauge is always on. 4 → gauge normally off. It turns on when the set point 1 is activated. If the set point 3 type (win 101) is selected as “pressure” the gauge is always on.

The windows from 136 to 140 manage an additional set point (set point 3) dedicated to turn ON or OFF the gauge (win 267=4)

Pump Driving

Typical pump driving chart start and normal operation with low or no gas load.

Stop Speed Reading and Active Stop

These functions can be activated by the Front Panel or serial line (Win 107 = Active Stop, Win 167 = Stop Speed Reading).

Active Stop brakes the pump using the motor. The Stop Speed reading reads the actual pump speed after a stop command.

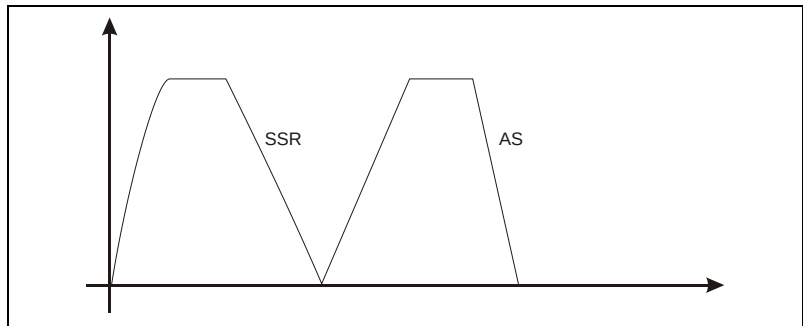


Figure 24 Diagram 5

J2 – Serial Connector

This connector provides the connection for RS-232 and RS-485 serial line.

J2 Pin-out:

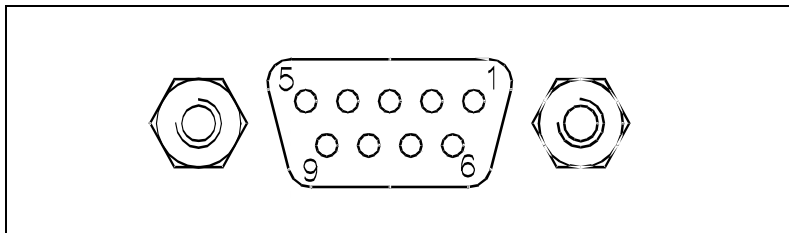


Figure 25 J2 Connector

Tab. 6

PIN N.	SIGNAL NAME
1	+5 V (OUT) (Reserved)
2	TX (RS-232)
3	RX (RS-232)
4	NC
5	GND
6	A+ (RS-485)
7	NC
8	B- (RS-485)
9	RESERVED

Connector Examples

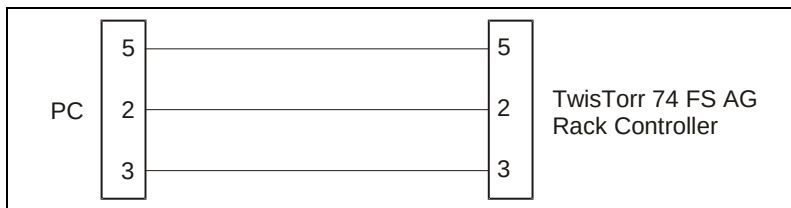


Figure 26 RS-232 Connection

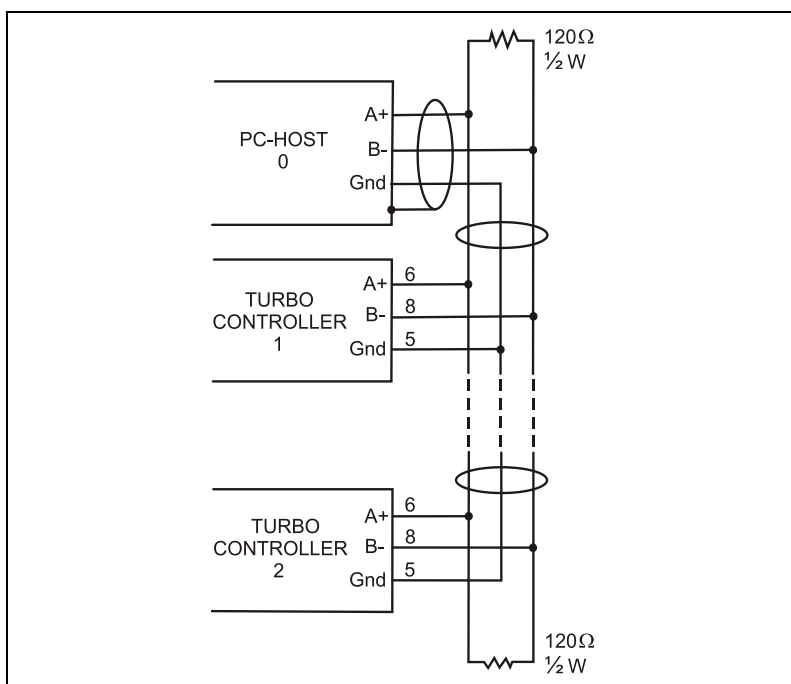


Figure 27 RS-485 Connection

The communication port mating connector is supplied with the RS-232 PCB (AMP/Cannon or equivalent 15-pin "D" type male connector). For example, the Transmit data signal from controller (pin 2) must be connected to the host computer's receive data line (pin 2) and vice versa. Consult the host computer instruction manual for its serial port connections.

NOTE

Agilent cannot guarantee compliance with FCC regulations for radiated emissions unless all external wiring is shielded, with the shield being terminated to the metal case on the D-subconnector. The cable should be secured to the connector with screws.

Serial Communication Descriptions

This unit can communicate by two different protocols:

- protocol (old system) "letter"
- "Window" protocol (new system)

These two protocols can be used as well with RS-232 or RS-485 media.

NOTE

Please use "Window" protocol for new development.

Letter Protocol Description

The default protocol is "Window"

Communication format:

- 8 data bit
- no parity
- 1 stop bit
- The baud rate is programmable via front panel from 600 to 9600 baud. The controller is factory-set for 9600 baud operation.

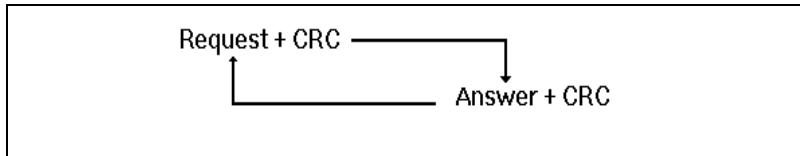
Communication protocol:

Host = Master

Controller = Slave

The communication is performed in the following way:

Host Controller



Request is an ASCII character identifying the action that must be performed by the controller or the requested information.

Allowed ASCII characters:

- "A" = START
- "B" = STOP
- "C" = Low Speed ON
- "D" = Low Speed OFF
- "E" = Request for operational parameters
- "F" = Pump times zeroing

17 Technical Information

Letter Protocol Description

- "G" = Parameters reading
- "H" = Parameters writing
- "I" = Request for operating status
- "J" = Request for numerical reading
- "K" = Request for counters reading.

CRC corresponds to the sum (with inverted sign) of all the preceding bytes.

e.g., the START command "A" in ASCII code = 41 inverted it will be: $FF + 1 - 41 = BF$.

Tab. 7

CHARACTER	ASCII	CONVERTED ASCII FOR CRC CALCULATION
"A"	41h	-
CRC	-	BFh
"B"	42h	-
CRC	-	BEh
"C"	43h	-
CRC	-	BDh
"D"	44h	-
CRC	-	BCh
"E"	45h	-
CRC	-	BBh
"F"	46h	-
CRC	-	BAh
"G"	47h	-
CRC	-	89h
"H"	48h	-
"I "	49h	-
CRC	-	B7h
"J"	4Ah	-
CRC	-	B6h

CHARACTER	ASCII	CONVERTED ASCII FOR CRC CALCULATION
"K"	4Bh	-
CRC	-	B5h
"ACK"	06h	-
CRC	-	FAh
"NACK"	15h	-
CRC	-	EBh

Answer = after a request from the host, the Controller will answer in one of the following ways:

- ACK
- NACK
- Message

When the Request is "A", "B", "C", "D", "F" the Controller will Answer the ACK or NACK.

When the Request is "E", the **Answer** will contain the complete set of the following parameters:

Tab. 8

BYTES	MEANING
1 MSB LSB - - X X X 	0 = STOP 1 = WAITING INTERLOCK 2 = STARTING 3 = NORMAL OPERATION 4, 5 = HIGH LOAD 6 = FAILURE 7 = APPROACHING LOW SPEED
2 ÷ 5	Cycle Time
6 ÷ 9	Pump Life
10 ÷ 11	Pump Temperature
12	Current*

17 Technical Information

Letter Protocol Description

BYTES	MEANING
13	Voltage*
14 ÷ 17	Frequency
18 ÷ 19	Cycle #
20	R1 Status
21	R2 Status
22	CRC

*The values for current and voltage are given as numbers, scaled from 0 to 255, where 0 corresponds to 0 V and 255 to the full scale voltage (130 V) or current (2,5 A).

When the **Request** is "G", the **Answer** will contain a string of 11 characters with the following parameters:

Tab. 9

BYTES	MEANING
1-2	Pump cycles number (integer coded in 2 bytes)
3	Speed threshold
4-7	Run up time in seconds (long coded in 4 bytes)
8	Deat time (0 = NO 1 = YES)
9	Reserved
10	Soft Start mode (0 = NO 1=YES)
11	CRC

When the configuration parameters have to be changed, send a **Request** string with "H" characters of the following type:

"H" + data + CRC

The following parameters can be changed:

Tab. 10

BYTES	MEANING
1	Speed threshold
2-5	Run up time in seconds (long coded in 4 bytes)
6	Deat time (0 = NO 1 = YES)
7	Reserved
8	Soft Start mode (0 = YES 1 = NO)
9	CRC

When the **Request** is "I" (Status readings), the **Answer** will contain a string of 2 characters with the following parameters:

Tab. 11

BYTES	MEANING
1	
MSB LSB	
--XXXXXX	
	0 = STOP
	1 = WAITING INTERLOCK
	2 = STARTING
	3 = NORMAL OPERATION
	4, 5 = HIGH LOAD
	6 = FAILURE
	7 = APPROACHING LOW SPEED
	R2 status 0 = OFF 1 = ON
	R1 status 0 = OFF 1 = ON
2	CRC

When the **Request** is "J" (Numerical readings), the **Answer** will contain a string of 5 characters with the following parameters:

Tab. 12

BYTES	MEANING
1	Current (0-255 scaled)
2	Voltage (0-255 scaled)
3	Rotational speed KRPM
4	Pump temperature °C (0-254 temperature reading, 255 = fail)
5	CRC

When the **Request** is "K" (Counters readings), the **Answer** will contain a string of 11 characters with the following parameters:

Tab. 13

BYTES	MEANING
1-4	Cycle time
5-8	Pump life
9-10	Cycle number
11	CRC

On request a sample program in QBasic language is available by Agilent.

Window Protocol

Description

Communication Format

- 8 data bit
- no parity
- 1 stop bit
- baud rate: 600/1200/2400/4800/9600 programmable

Communication Protocol

The communication protocol is a MASTER/SLAVE type where:

- Host = MASTER
- Controller = SLAVE

The communication is performed in the following way:

- 1 the host (MASTER) send a MESSAGE + CRC to the controller (SLAVE);
- 2 the controller answer with an ANSWER + CRC to the host.

The MESSAGE is a string with the following format:

<STX>+<ADDR>+<WIN>+<COM>+<DATA>+<ETX>+<CRC>

Where:

NOTE

When a data is indicated between two quotes ('...') it means that the indicated data is the corresponding ASCII character.

When a number is indicated with 0x prefix it means it is an hexadecimal number (16 base coded).

<STX> (Start of transmission) = 0x02

<ADDR> (Unit address) = 0x80 (for RS 232)

<ADDR> (Unit address) = 0x80 + device number (0 to 31)
(for RS 485)

<WIN> (Window) = a string of 3 numeric character indicating the window number (from '000' to '999'); for the meaning of each window see the relevant paragraph.

<COM> (Command) = 0x30 to read the window, 0x31 to write into the window

<DATA> = an alphanumeric ASCII string with the data to be written into the window. In case of a reading command this field is not present. The field length is variable according to the data type as per the following table:

Tab. 14

Data Type	Field Length	Valid Characters
Logic (L)	1	'0' = OFF '1' = ON
Numeric (N)	6	'-', '.', '0' ... '9' right justified with '0'
Alphanumeric (A)	10	from blank to '_' (ASCII)

<ETX> (End of transmission) = 0x03

<CRC> = XOR of all characters subsequent to <STX> and including the <ETX> terminator. The value is hexadecimal coded and indicated by two ASCII character.

The addressed SLAVE will respond with an ANSWER whose structure depends from the MESSAGE type. When the MESSAGE is a reading command, the SLAVE will respond transmitting a string with the same structure of the MESSAGE.

NOTE

Using the RS 485 interface, the message structure remains identical to the one used for the RS 232 interface, the only difference being that the value assigned to the ADDRESS <ADDR>

The controller can answers with the following response types:

Tab. 15

Response Type	Response Length	Response Value	Description
Logic	1 byte	-	after a read instruction of a logic window
Numeric	6 bytes	-	after a read instruction of a numeric window
Alphanumeric	10 bytes	-	after a read instruction of an alphanumeric window
ACK	1 byte	(0x6)	the command execution has been successfully completed
NACK	1 byte	(0x15)	the command execution has been failed
Unknown Window	1 byte	(0x32)	the specified window in the command is not a valid window
Data Type Error	1 byte	(0x33)	the data type specified in the command (Logic, Numeric or Alphanumeric) is not accorded with the specified Window
Out of Range	1 byte	(0x34)	the value expressed during a write command is out of the range value of the specified window
Win Disabled	1 byte	(0x35)	the specified window is Read Only or temporarily disabled (for example you can't write the Soft Start when the Pump is running)

Examples

Command: START

Source: PC

Destination: Controller

02	80	30	30	30	31	31	03	42	33
STX	ADDR	WINDOW			WR	ON	ETX	CRC	

Source: Controller

Destination: PC

02	80	06	03	38	35
STX	ADDR	ACK	ETX	CRC	

Command: STOP

Source: PC

Destination: Controller

02	80	30	30	30	31	30	03	42	32
STX	ADDR	WINDOW			WR	OFF	ETX	CRC	

Source: Controller

Destination: PC

02	80	06	03	38	35
STX	ADDR	ACK	ETX	CRC	

Command: SOFT-START (ON)

Source: PC

Destination: Controller

02	80	31	30	30	31	31	03	42	32
STX	ADDR	WINDOW			WR	ON	ETX	CRC	

Source: Controller

Destination: PC

02	80	06	03	38	35
STX	ADDR	ACK	ETX	CRC	

Command: SOFT-START (OFF)

Source: PC

Destination: Controller

02	80	31	30	30	31	30	03	42	33
STX	ADDR	WINDOW			WR	OFF	ETX	CRC	

Source: Controller

Destination: PC

02	80	06	03	38	35
STX	ADDR	ACK	ETX	CRC	

17 Technical Information

Examples

Command: READ PUMP STATUS

Source: PC

Destination: Controller (with address = 3)

02	83	32	30	35	30	03	38	37
STX	ADDR	WINDOW			RD	ETX	CRC	

Source: Controller (with address = 3 in stop status)

Destination: PC

02	83	3	3	3	3	3	3	3	3	3	3	03	3	3
		2	0	5	0	0	0	0	0	0	0		8	7
STX	ADDR	WINDOW			R D	DATA (STATUS)						ETX	CRC	

Command: READ SERIAL TYPE

Source: PC

Destination: Controller (with address = 3 in 485 mode)

02	83	35	30	34	30	03	38	31
STX	ADDR	WINDOW			RD	ETX	CRC	

Source: Controller

Destination: PC

02	83	35	30	34	30	31	03	42	30
STX	ADDR	WINDOW			RD	DATA	ETX	CRC	

Window Meanings

Tab. 16 Accessories and Spare Parts

N.	Read/ Write	Data Type	Description	Admitted Values
000	R/W	L	Start/Stop (in remote/ Front mode the window is a read only)	Start = 1 Stop = 0
008	R/W	L	Remote (default) or Serial configuration	Remote = 1 Serial = 0 (default = 1)
100	R/W	L	Soft Start (write only in Stop condition)	YES = 1 NO = 0
101	R/W	N	R1 Set Point type	0 = Frequency 1 = Power 2 = Time 3 = Normal (default = 3) 4 = Pressure (available only if the gauge is connected)
102	R/W	N	R1 Set Point threshold value (expressed in Hz, W or s)	(default = 900)
103	R/W	N	Set Point delay: time between the pump start and the set point check (seconds)	0 to 99999 (default = 0)
104	R/W	L	Set Point signal activation type: the signal can be "high level active" or "low level active"	0 = high level active 1 = low level active (default = 0)
105	R/W	N	Set point hysteresis (in % of value)	0 to 100 (default = 2)
106	R/W	L	Water cooling	0 = NO 1 = YES
107	R/W	L	Active Stop (write only in stop)	0 = NO 1 = YES
108	R/W	N	Baud rate	600 = 0 1200 = 1 2400 = 2 4800 = 3 9600 = 4 (default = 4)
109	W	L	Pump life/ cycle time/ cycle number	To reset write '1'

17 Technical Information

Examples

reset				
110	R/W	L	Interlock type (default = 0)	Impulse = 0 Continuous = 1
111	R/W	L	Analog output type: output voltage signal proportional to frequency, power, temperature or pressure	0 = frequency 1 = power (default = 1) 2 = Pump temperature 3 = Pressure
120	R/W	N	Rotational frequency setting (Hz)	Low speed frequency (win 117) to 1167 (default = 1167)
122	R/W	L	Set vent valve on/off (on = closed)	On = 1 Off = 0 (default = 1)
123	Reserved to Agilent service			
124				
125	R/W	L	Set the vent valve operation	Automatic = 0 (see note 1.) On command = 1 (see note 2.) (default = 0)
126	R/W	N	Vent valve opening delay (expressed in 0.2 sec)	0 to 65535 (corresponding to 0 to 13107 sec)
130	Reserved to Agilent service			
136	R/W	N	Gauge Set Point Type	0 = Freq 1 = Power 2 = Time 3 = Normal (default)
137	R/W	N	Gauge Set Point Value (Hz, W, s)	(default (867)
138	R/W	N	Gauge Set Point Mask (sec)	(default = 0)
139	R/W	L	Gauge Set Point Segnal Activation Type	0 = high level active (default) 1 = low level active
140	R/W	N	Gauge Set front Mysteresis (in % of R2 Valve)	(default = 2)
143	R/W	N	External Fan Configuration	0 = ON 1 = automatic 2 = serial (default = 0)

144	R/W	L	External Fan Activation	0 = OFF 1 = ON (default = 0)
147	R/W	N	Vent open time See "vent connector" paragraph	0 = infinite 1 bit = 0.2 sec
155	R	N	Power limit applied Read the maximum power deliverable to the pump	watt
157	R/W	N	Gas load type. Select the gas load to the pump	0 = N ₂ 1 = Argon
162	R/W	A	R1 Set Point Pressure Threshold Valid if min. 101 = 4 Format X.X EsXX Where X = 0 to 9 s = + or -	
163	R/W	N	Pressure unit of measure	0 = mBar 1 = Pa 2 = Torr
167	R/W	A	Stop speed reading Activates / deactivates the pump speed reading after Stop command	0 = disable 1 = enable
171	R/W	N	R2 Set Point Type	0 = Freq 1 = Power 2 = Time 3 = Normal (default = 3) 4 = Pressure (available only if the gauge is connected)
172	R/W	N	R2 Set Point Value (Hz, W, s)	
173	R/W	N	R2 Set Point Mask (sec)	
174	R/W	L	R2 Set Point Signal Activation Type	0 = high level active 1 = low level active
175	R/W	N	R2 Set Point hysteresis (in % of R2 Valve)	
176	R	A	R2 Set Point Pressure Threshold Valid if win 171 = 4 Format X.X EsXX Where: X= 0 to 9 s = + or -	
177	R/W	L	Start Output Mode	0 = Starting (Output ON only with pump Status = Starting) 1 = running (Output ON when the pump is running) (default -0)

17 Technical Information

Examples

181	R/W	N	Gas type	0 = not configured 1 = Nitrogen 2 = Argon 3 = Hydrogen 4 = other
182	R/W	N	Gas correction	
200	R	N	Pump current in mA dc	
201	R	N	Pump voltage in Vdc	
202	R	N	Pump power in W (pump current x pump voltage duty cycle)	
203	R	N	Driving frequency in Hz	
204	R	N	Pump temperature in °C	0 to 70
205	R	N	Pump status	Stop = 0 Waiting intlk = 1 Starting = 2 Auto-tuning = 3 Braking = 4 Normal = 5 Fail = 6
206	R	N	Error code	Bit description: see the following figure
211	R	N	Controller Heatsink Temperature (°C)	
216	R	N	Controller Air Temperature (°C).	
224	R	A	Pressure reading Format = X.X E XX	
226	R	N	Rotation Frequency (rpm)	
257	R	N	Gauge status	0 = No gauge connected 1 = gauge connected 2 = Under range/gauge error 3 = over range/gauge error 4 = Rid unknown
267	R/W	N	Gauge power	0 = gauge off 1 = gauge On 2 = programmable power sp1 3 = programmable power sp2 4 = programmable power sp3
300	R	N	Cycle time in minutes (zeroed by the reset command)	0 to 999999
301	R	N	Cycle number (zeroed by the reset command)	0 to 9999

302	R	N	Pump life in hours (zeroed by the reset command)	0 to 999999
320 to 399	Reserved to Agilent service			
400	R	A	CRC EPROM (QE)	QE8XXXX (where "XXXX" are variable)
402	R	A	CRC Param. (PA)	PA8XXXX (where "XXXX" are variable)
404	R	A	CRC Parameter structure	"XXXX"
500	Reserved to Agilent service			
503	R/W	N	RS 485 address	0 to 31 (default = 0)
504	R/W	L	Serial type select	0 = RS 232 1 = RS 485 (default = 0)

NOTE

1. Automatic means that when the controller stops, the vent valve is opened with a delay defined by window n. 126; when the controller starts, the vent valve is immediately closed.
2. On command means that the vent valve is opened or closed by means of window n. 122.

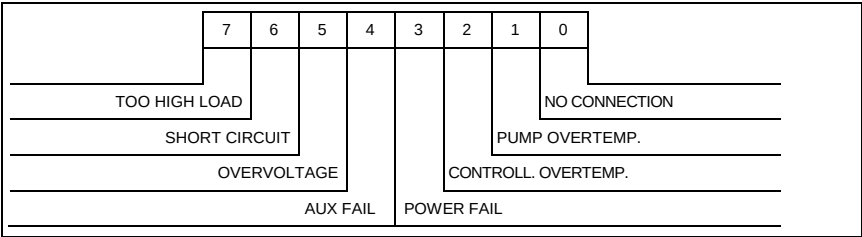


Figure 28 Window N. 206 Bit Description

How to Use by Front Panel

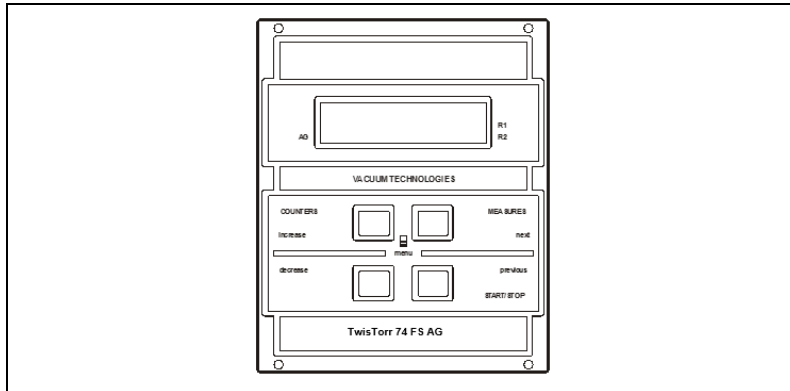


Figure 29

- **Start Stop and Reset Fail button**

Press once for the START command and again for the STOP command. In the case of controller's failure, press once to reset the fail condition.

NOTE

The front panel START/STOP function is available only if the controller is set to "FRONT mode". If it is set to "REMOTE mode", when the key is pressed an error message is shown indicating "COMMAND LOCKED IN REMOTE", if it is set to "SERIAL" mode, the "COMMAND LOCKED IN SERIAL MODE" error message is displayed.

The "Reset Fail" function is always available regardless of controller mode.

- Counters**
Pressing the key once, the following information is displayed:

			C	Y	C	L	E		X	X	X	X			
		T	I	M	E			X	X	X	X	X	m		

Where:

CYCLE XXXX Number of start/stop cycles

TIME XXXXXm Time of last cycle (minutes)

Pressing twice, the following is displayed:

			P	U	M	P		L	I	F	E				
				X	X	X	X	X	X	h					

PUMP LIFE

XXXXXX total operating time (hours) of the pump

Pressing a third time, the following message is displayed:

	M	N	:		X	X	X	X	X	X	X	X	X	
	S	N	:		Y	Y	Y	Y	Y	Y	Y	Y	Y	

MN Model Number of the controller

SN Serial Number

By pressing the key again, the display returns back to the main screen.

- Measures**
Pressing once, the following is displayed:

P	=	X	X	X	W					X	X	X	X	H	z
					T	=	X	X	°	C					

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How to Use by Front Panel

Where:

P = XXX W is the current power supplied to the pump

T = XX °C is the current temperature of the pump

XXXX Hz is the driving frequency of the pump. This value is set by default to 1167 Hz but can be modified by the user according to needs, proceeding as described in the “PROGRAMMING” paragraph. The unit of measurement (HZ / Krpm, default = Hz) is selected by the user from the INPUT/OUTPUT->SPED menu.

If the SOFT-START function (PUMP SETTING menu → SOFTSTART) is active and the pump is ramping, indication of the speed of rotation is replaced with a bargraph that indicates the soft-start step in course and the total number of steps to be performed. On completion of the soft-start ramp, the bargraph is replaced with indication of the frequency of rotation of the pump

Pressing twice, the following is obtained:

N	O	R	M	A	L					X	X	X	X	H	z
	I	=	X	.	X	X	A		V	=	X	X	X	V	

Where:

NORMAL: indicates the pump status

XXXXHz: is the driving frequency of the pump (for further details please see the description “Pressing once” the **Measures** key)

I=X.XXA: indicates the current absorbed by the pump

V= XXX V: indicates the pump driving voltage

Pressing 3 times, the following is obtained:

P	=	X	X	X	W					X	X	X	X	H	z
		X	.	X	X	E	-	Y	Y	m	B	a	R		

X	.	X	E	-	Y	Y	m	B	a	r		Z	Z	Z	Z
	P	=	X	X	X	W			X	X	X	X	H	z	

Where:

P= XXXW: indicates the current power supplied to the pump

XXXHz: is the driving frequency of the pump (for further details please see the description “Pressing once” the **Measures** key)

X.XE-YYmBar: indicates the pressure currently measured by the gauge.

ZZZZ: indicates the kind of gauge (see table of gauge supported in the gauge chapter)

Other possible indications:

P	=	X	X	X	W					X	X	X	X	H	z
		G	A	U	G	E		E	R	R	O	R			

Where:

GAUGE_ERROR: gauge not connected or broken

P	=	X	X	X	W					X	X	X	X	H	z
	P	R	E	S	S	U	R	E		U	N	D	E	R	

Where:

PRESSURE UNDER: indicates that the pressure read is below the minimum measurement threshold set.

P	=	X	X	X	W					X	X	X	X	H	z
	P	R	E	S	S	U	R	E		O	V	E	R		

Where:

PRESSURE OVER: indicates that the pressure measured is above the maximum measurement threshold set.

The unit of measurement of the pressure can be selected (GAUGE menu→Unit) from “mBar”, “Pa”, “torr”

Pressing 4 times returns to the following screen page:

N	O	R	M	A	L					X	X	X	X	H	z
	X	X	X	X	X	X	X	X	X	X	X	X	X	X	

17 Technical Information

How to Use by Front Panel

Where:

NORMAL: indicates the operating regime of the pump

XXXHz: is the driving frequency of the pump (for further details please see the description “Pressing once” the **Measures** key)

The operating mode of the controller is shown at the bottom in the center. This may be:

FRONT: front mode (the controller accepts START/STOP and LOW SPEED commands from the front panel)

REMOTE: remote mode (the controller accepts START/STOP and LOW SPEED commands via the rear connector)

RS 232: the controller accepts commands from the serial port according to the RS 232 standard.

RS-485 [XX]: the controller accepts commands from the serial port in RS-485 mode with address XX (XX from 0 to 31)

PROFIBUS [YYY]: the controller accepts commands from the PROFIBUS port with address YYY

Black rectangles as shown below may be displayed at any time in the first and last box of the bottom line of the display (opposite the AG / R1 / R2 captions). The meaning of these rectangles is as follows:

AG → the Gauge is connected to the controller

R1 → relay (set point) R1 is active

R2 → relay (set point) R2 is active

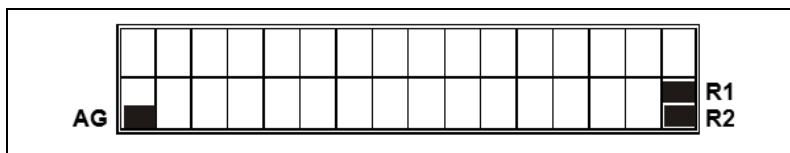


Figure 30

Use

General

Make all vacuum manifold and electrical connections and refer to TwisTorr pump instruction manual before to operating the TwisTorr controller.

WARNING!

Refer to the relevant pump manual to operate it in a safe way before starting.



NOTE

The input signal J1 interlock should be left in position including the shipping links if no external connections are made. The forepump and the TwisTorr pump can be switched on at the same time.

Startup

- Plug the controller power cable into a suitable power source.
The display lights up for approx. 2 seconds and shows:

				A	U	T	O	T	E	S	T				
							O	K							

- After 2 seconds, the display shows the following screen page:

S	T	O	P							.	.	.	.	H	z
					F	R	O	N	T						

NOTE

If the pump is not connected, the display will be as follows:

C	H	E	C	K		C	O	N	N	E	C	T	I	O	N
			T	O		P	U	M	P						

▪

- Pressing the START button, the pump is started and the display shows the following screen page.

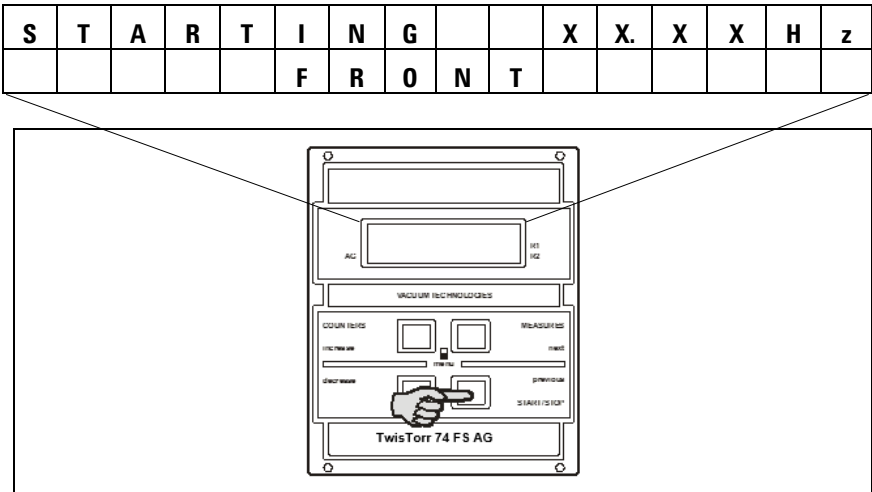


Figure 31

This screen page will be visible throughout the acceleration phase the pump. The increasing value of the rotational frequency will be displayed in the boxes marked X; the user may decide to express this value in Hz (default) or in KRMP.

Once operating speed has been reached, the following screen page is displayed:

N	O	R	M	A	L					1	1	6	7	H	z
					F	R	O	N	T						

17 Technical Information

Use

- Press the MEASURES push button: the display shows::

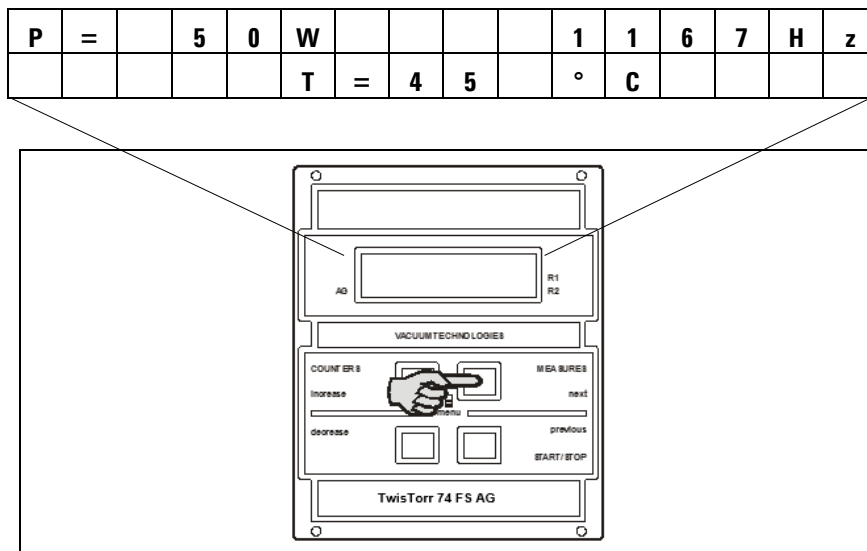


Figure 32

Where:

P = is the DC power drawn by the pump (controller display range 0 to 999 Watt)

Hz = is the rotational speed of the pump as a function of the controller output frequency

°C = is the temperature of the outer ring of the upper bearing (controller display range 00 to 99 °C)

- Pressing MEASURES key twice, the following screen page is displayed:

N	O	R	M	A	L					1	3	5	0	H	z
	I	=	1	.	2	0	A		V	=		4	2	V	

Where:

I is the current supplied by the controller to the pump at that precise moment.

V is the voltage at which the controller is powering the pump at that moment.

- Pressing the MEASURES key three times, with the GAUGE connected and functioning, the following screen page is displayed:

7	.	4	E	-	0	5	m	B	A	r		X	X	X	X
	P	=		5	0	W			1	3	5	0	H	z	

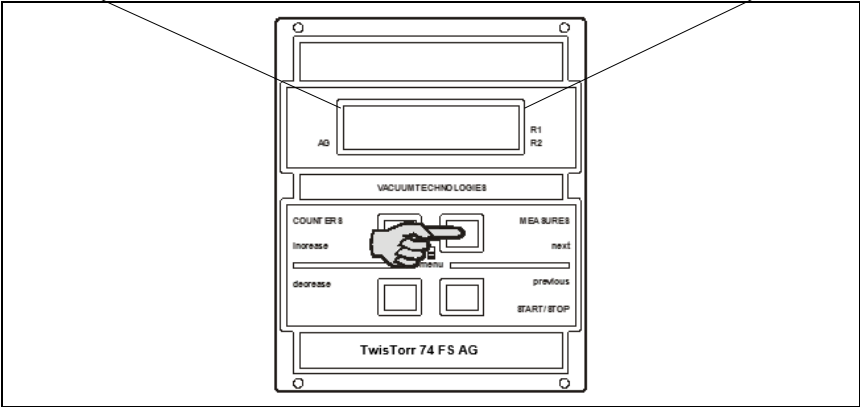


Figure 33

Where:

mBar shows the pressure value measured

XXXX shows which gauge is connected

The following messages may be displayed instead of the pressure:

GAUGE ERROR: indicates that the GAUGE is not identified

PRESSURE UNDER: indicates that a pressure is below the minimum threshold

PRESSURE OVER: indicates that a pressure is exceeding the maximum threshold

- Pressing the COUNTERS key, the following is obtained:

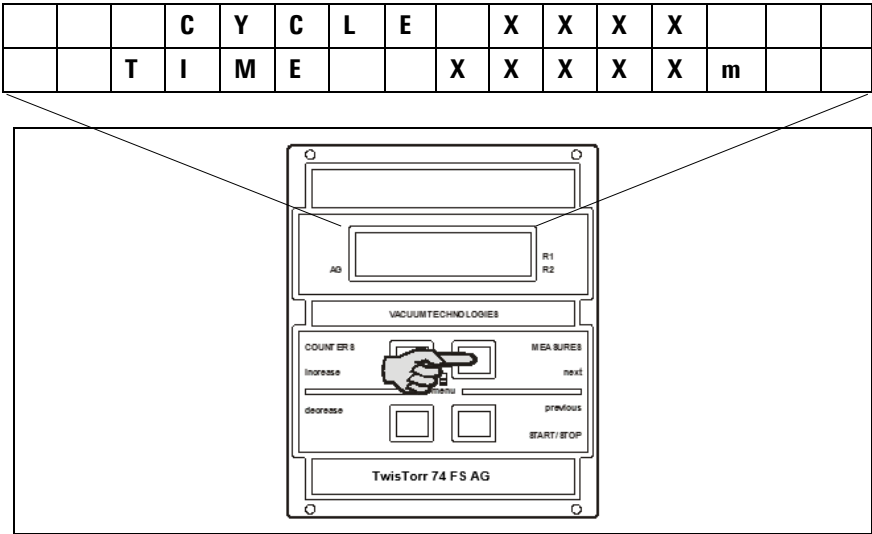


Figure 34

Where:

CYCLE XXXX
TIME XXXXXm

Number of start/stop cycles
Time of last cycle (minutes)

Pressing again:

			P	U	M	P		L	I	F	E				
				X	X	X	X	X	X	h					

Where:

PUMP LIFE
XXXXXX

total operating time (hours) of the pump

Programming

Configuration Menu

To access the configuration menu, press the COUNTERS + MEASURES buttons at the same time for at least 2 sec.

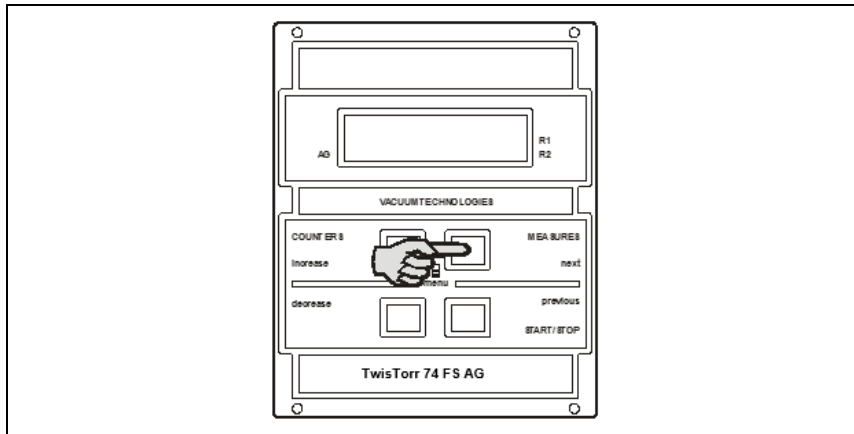


Figure 35

In programming mode, the meaning of the 4 buttons changes as follows:

- **COUNTERS** becomes “**INCREASE** and **DECREASE**” and makes it possible to select the next or previous value of a parameter (in the case of compulsory parameters) or to increase/decrease the individual digit of a parameter (in the case of numeric parameters)
- The “**MEASURES** and **START/STOP**” buttons become “**NEXT** and **PREVIOUS**” and make it possible to select the next parameter (or next digit of a numeric parameter) or previous parameter.

Switching from one parameter to the next (using the “**NEXT**” key) indicates confirmation of any changes made to the specific parameter. If the changes cannot be accepted by the controller, the “**Data out of limits**” error message is displayed, the changes are cancelled and the value of the parameter prior to the modification is repropose.

Switching from a parameter to the previous parameter (using the “**PREVIOUS**” key) indicates the intention to abort any changes made to the specific parameter. Abort is indicated by the caption “**Changes aborted**” that is displayed for approx. 1 second before moving to the previous parameter.

The configuration environment has a tree-like structure and permits “circular” navigation; on reaching the end of a branch, the user is returned automatically to the start of this. Navigation can proceed in a forwards (**NEXT** button) or backwards (**PREVIOUS** button) direction

It is possible to enter a sub-branch by pressing the “**INCREASE**” button. If no buttons are pressed for more than 1 minute, the controller automatically quits the programming environment canceling any modifications made to the current parameter.

The figure below shows the first level of the configuration menu and the links to the respective second level menus (indicated below).

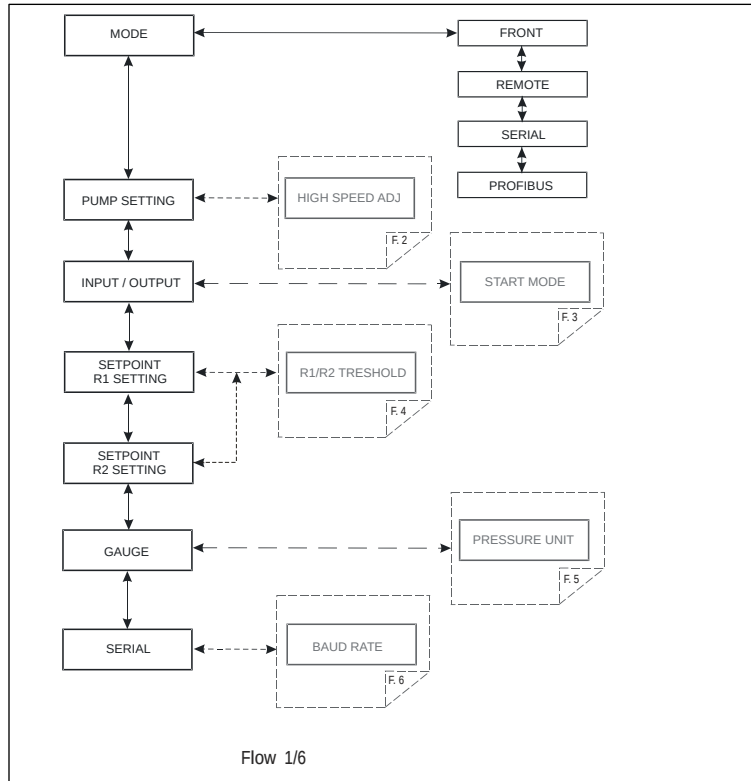


Figure 36 Menu structure

Mode Menu

- **MODE:** permits selection (see Flow 1/6) of the operating mode of the controller from those indicated below:
 - **FRONT:** accepts commands from the front panel
 - **REMOTE:** accepts commands from the remote input connector
 - **SERIAL:** accepts commands from the serial interface (or Profibus interface if present)
 - **PROFIBUS:** this operating mode is present only if the Profibus interface is installed on the controller

Pump Setting Menu

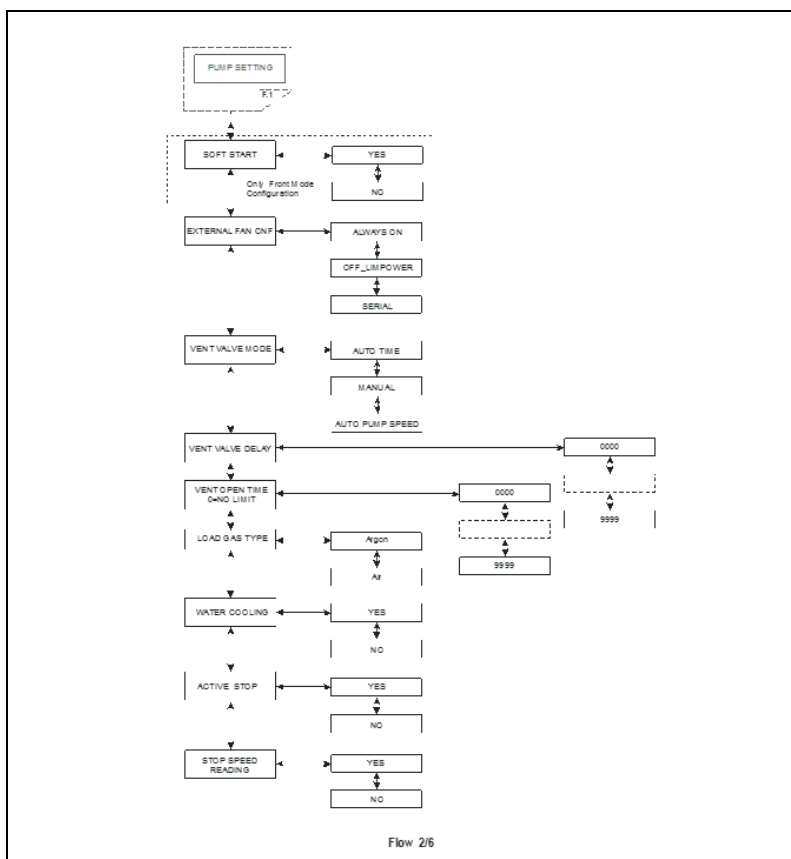


Figure 37 Pump setting menu

This menu contains all the parameters that have a direct effect on driving of the pump.

- **SOFT START:** enables (YES) or disables (NO) the soft start function. This parameter is visible only if the controller is configured in FRONT mode.

- **EXTERNAL FAN CNF:** Makes it possible to select driving mode of the external cooling fan. Possible options:
 - **ALWAYS ON:** the cooling fan is always ON
 - **OFF_LIMPOWER:** 10 seconds after the pump has reached NORMAL status, the fan is switched off and the driving power of the pump is reduced to 50 W
 - **SERIAL:** the cooling fan is controlled directly via the serial port.
- **VENT VALVE MODE:** makes it possible to select driving of the vent valve
 - **AUTO TIME:** the valve is controlled by the controller
 - **MANUAL:** the valve is controlled by the serial port.
 - **AUTO PUMP SPEED:** the valve is controlled by the controller its king into account the pump's deceleration.
- **VENT VALVE DELAY:** Sets the delay, expressed in seconds, between stopping of the pump and opening of the vent valve.
- **VENT VALVE OPEN TIME:** sets the time, expressed in seconds, of opening of the vent valve. Enter 0 if the valve is to be kept open indefinitely (until the next start).
- **LOAD GAS TYPE:** selects the type of gas used in the process. Therefore, the controller limits the power supplied to the pump according to the setting of this parameter.
- **WATER COOLING:** indicates whether the pump is cooled with water or not. The controller limits the power supplied to the pump according to this setting.
- **ACTIVE STOP:** enables (YES) or disables (NO) active braking.
- **STOP SPEED READING:** enables (YES) or disables (NO) the pump rotational frequency reading function also during the slowing phase that follows a stop command

Input/Output Menu

This menu contains parameters that determine the operating mode of the inputs and outputs of the controller.

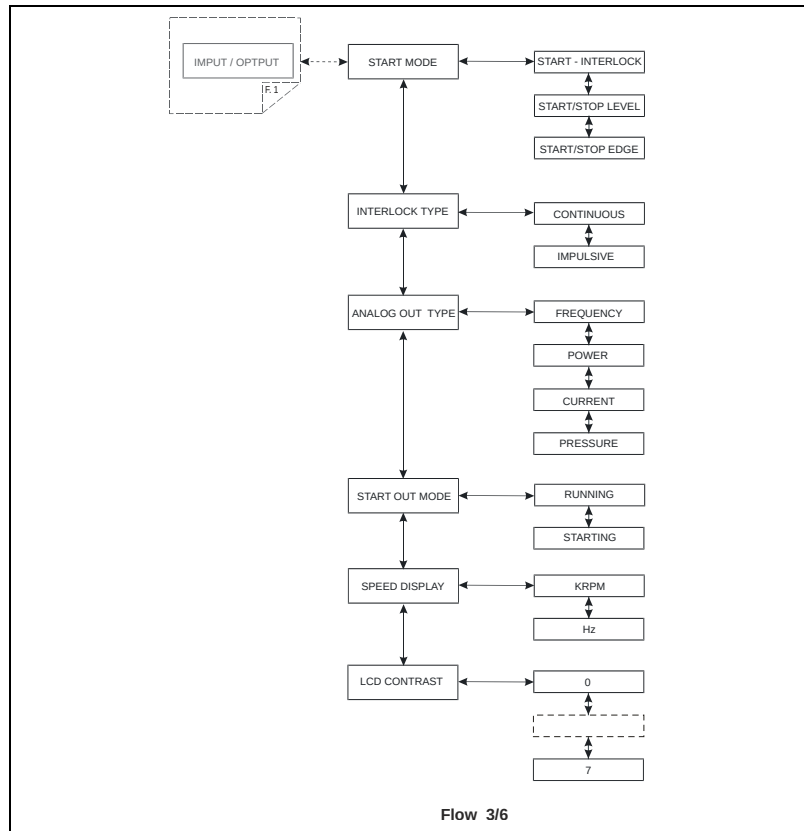


Figure 38 I/O menu

- **START MODE:** Permits selection of one of the three types of functioning of the START/STOP and INTERLOCK inputs:

- **START/INTERLOCK:** The pump is started if the START/STOP input is activated and is stopped if the START/STOP input is de-activated. The interlock input must be constantly active (INTERLOCK TYPE parameter = continuous) or may be active only at the time of starting (INTERLOCK TYPE parameter = impulsive)
- **START/STOP LEVEL** The START/STOP input acts as only START command (active on the level) The INTERLOCK input acts as STOP command (active on the level).
In the case of concurrent activation of the two inputs, the STOP command takes priority
- **START/STOP EDGE** The START/STOP input acts as START command (active on the edge)
The INTERLOCK input acts as STOP command (active on the edge).
- **INTERLOCK TYPE:** This parameter is displayed only when the previous parameter (START MODE) is equal to START/INTERLOCK and permits selection of one of the following two operating modes of the INTERLOCK input:
 - **CONTINUOUS:** the INTERLOCK input must be active continuously so that the pump can be driven by the controller. If, with the controller in start, the input INTERLOCK is de-activated, the controller switches to WAITING INTERLOCK status and interrupts driving of the pump
 - **IMPULSIVE:** it is sufficient for the input to be active when the START command is sent to the pump. Subsequently, the input can also be de-activated. If the interlock is not active when the pump is started, the controller switches too WAITING INTERLOCK status.

- **ANALOG OUT TYPE:** permits selection of the type of signal to be sent to the programmable analogue output. One of the following signals can be selected:
 - **FREQUENCY:** the drive frequency of the pump is shown
 - **POWER:** the power supplied to the pump is shown
 - **TEMPERATURE:** the temperature of the pump is shown
 - **PRESSURE:** the pressure read by the gauge is shown.
- **START OUT MODE:** Selects the operating mode of the START output. Possible options:
 - **RUNNING:** the output is active when the pump is driven
 - **STARTING:** the output is active only during acceleration ramp-up.
- **SPEED OUT MODE (KRPM/Hz):** permits selection of the unit of measurement of the speed of rotation shown on the display of the controller.
This setting acts only on the measurement shown on the display. Possible options:
 - **KRPM**
 - **Hz**
- **LCD CONTRAST:** selectable from 0 to 7.

Setpoint R1 and Setpoint R2 Submenus

This submenu contains the settings relating to the output with relay R1 and R2 (if it is connected a gauge it will present also the settings for the set point 3).

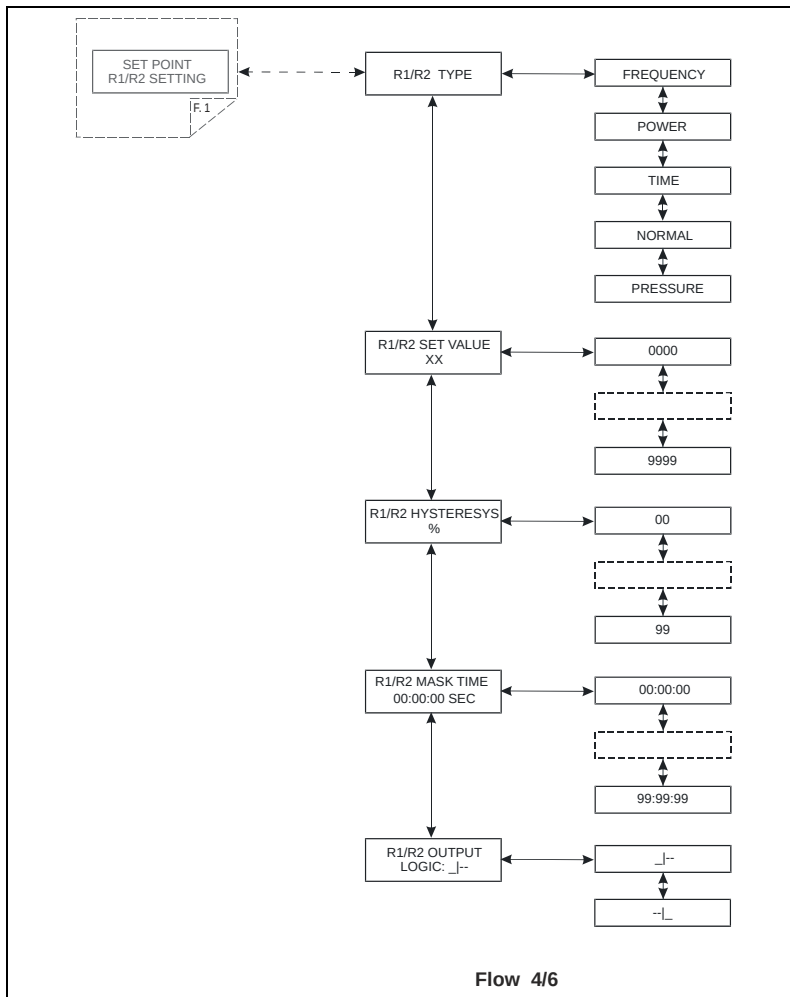


Figure 39 Setpoint menu

- **R1 (R2) TYPE:** selects the type of reference measurement to drive the relay. Possible types:
 - Frequency: driving frequency of the pump
 - Power: power supplied to the pump
 - Time: time from start
 - Normal: achievement of normal condition
 - Pressure: pressure read by the gauge, if present
- **R1 (R2) SET VALUE:** Sets the reference value at which the relay R1 (R2) is activated. This parameter has no effect if NORMAL (previous parameter) is selected as SETPOINT TYPE.
- **R1 (R2) HISTERESYS:** Sets, in percentage, the hysteresis of the relay
- **R1 (R2) MASK TIME:** sets the masked time (in hh:mm:ss), measured from start, within which the SETPOINT R1 (R2) is not driven even if the threshold value has been reached. This parameter is not available if SETPOINT TYPE is set to NORMAL
- **R1 (R2) OUTPUT LOGIC:** Selects the control logic of the SETPOINT output.
 - positive logic (the relay is energized when the measurement exceeds the reference)
 - negative logic (the relay is energized when the measurement is below the reference)

Gauge Menu

This menu contains the parameters relating to functioning of the active gauge.

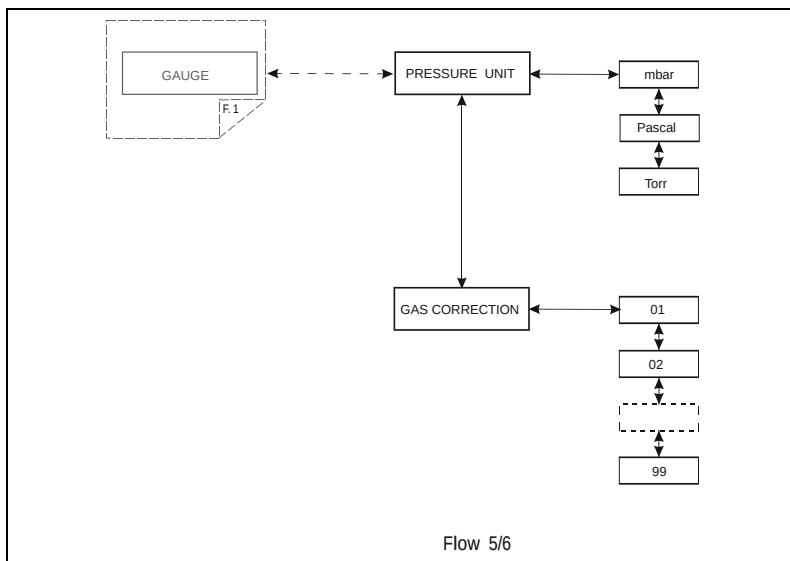


Figure 40 Gauge menu

- **PRESSURE UNIT:** selects the unit of measurement of the pressure used to show this on the display, for display on the serial line and for setting of the setpoint and autopressure thresholds. Possible options:
 - mbar
 - Pascal
 - Torr

GAUGE MODEL ID: it shows the gauge model identified by the controller.

If there are more model with the same identification the user can select the right gauge connected.

- **GAS TYPE :** sets the gas type:
 - 0 : not configured
 - 1 : Nitrogen
 - 2 : Argon
 - 3 : Idrogen
 - 4 : Other
- **CORRECTION :** if it is selected the gas type 4 (Other) you can set the correction factor of the pressure measurement for this kind of gas. The pressure indicated by the instrument is equal to:

$$P_{\text{indicated}} = P_{\text{measured}} \times \text{Gas correction} / 10$$

Serial Menu

This menu contains the settings for management of the RS-232/RS-485 serial line.

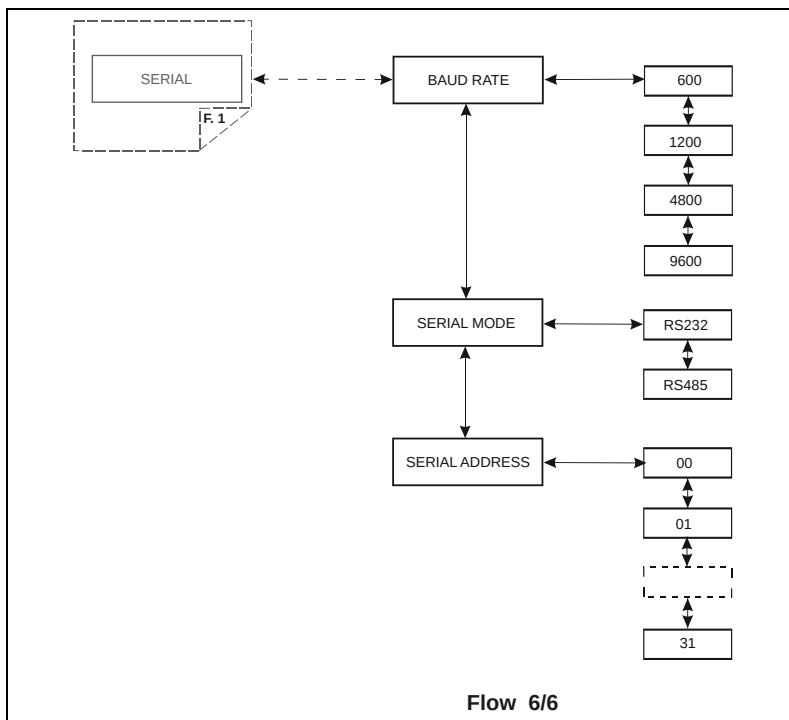


Figure 41 Serial menu

- **BAUD RATE:** permits selection of the communication speed selecting this from 600, 1200, 4800, 9600 baud
- **SERIAL MODE:** permits setting of the type of interface to RS-232 (point to point) or RS-485 (multidrop)
- **SERIAL ADDRESS:** sets the address of the node in the case of the RS-485 interface. This parameter is not visible if the RS-232 is selected.

How to Use by Remote I/O

- 1 Power ON the controller
- 2 Press the “COUNTERS” and “MEASURES” button together.
- 3 Select MODE = REMOTE from the “INPUT/OUTPUT” menu
- 4 Exit from configuration menu
- 5 Connection example: (see Fig. 44)

How to Use in Serial Mode

- 1 Power on the controller
- 2 Connect the serial cable
- 3 Select by front panel if you want to work in 232 or 485, and if necessary, the serial address.
- 4 Select SERIAL from the INPUT/OUTPUT menu
- 5 Open the T-PLUS software (see orderable parts table)
- 6 Ready to operate by SERIAL connections.

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How to Use in Serial Mode

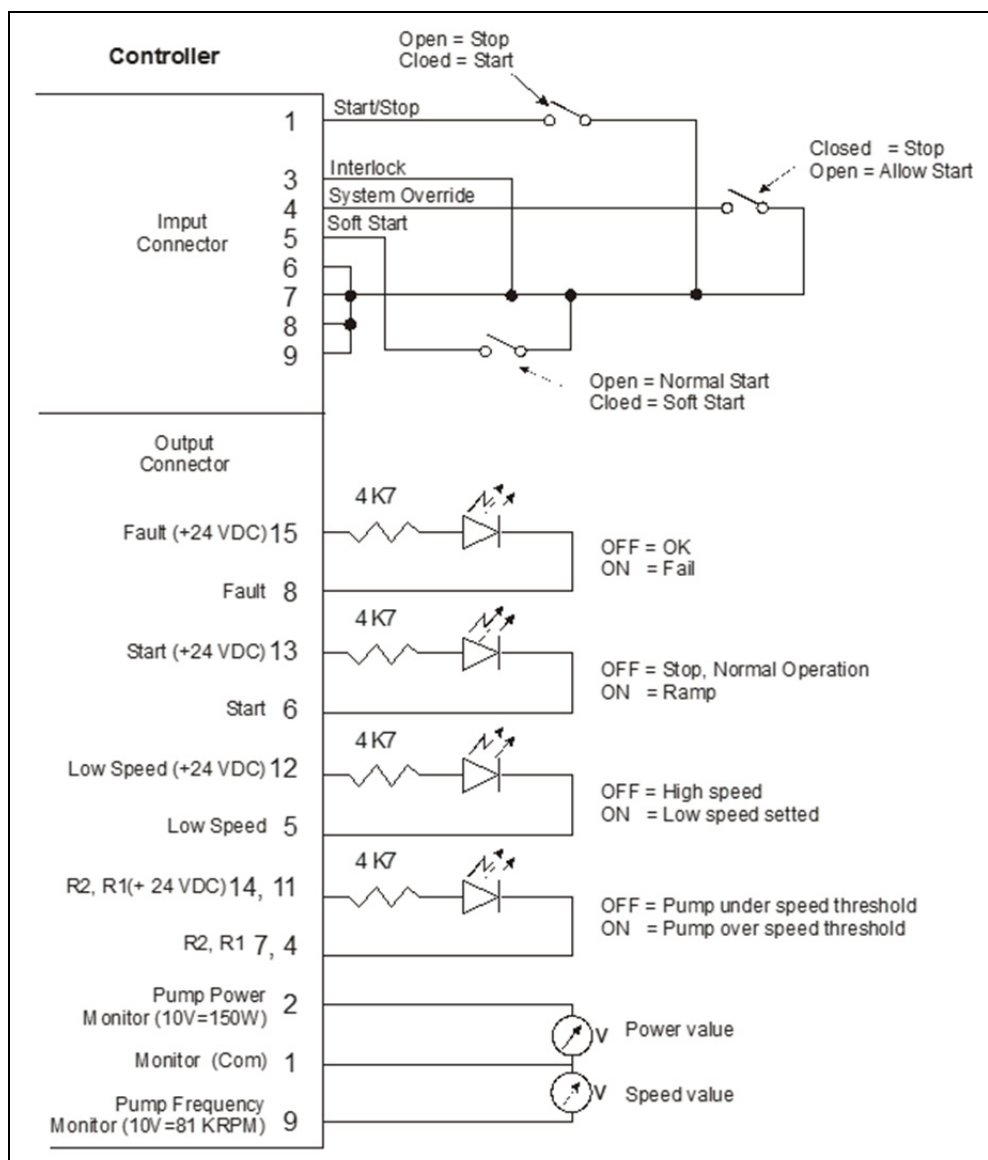


Figure 42 Connection example

Profibus Option

General

The Profibus option allow to communicate with the controller as Profibus DP slave device.

The option works as a bi-directional bridge between the Profibus network and the Agilent turbo controller internal link. The interface provides input data (measures) and output data (commands) to the controller, and in addition provides standard and user related diagnostics, in order to allow the user to manage the exceptions.

Installation

This is a factory installed option on controllers with the part number X3508-64022.

The Profibus connector pin-out is reported in the following figure. Please in order to obtain best performance use Profibus certified cables.

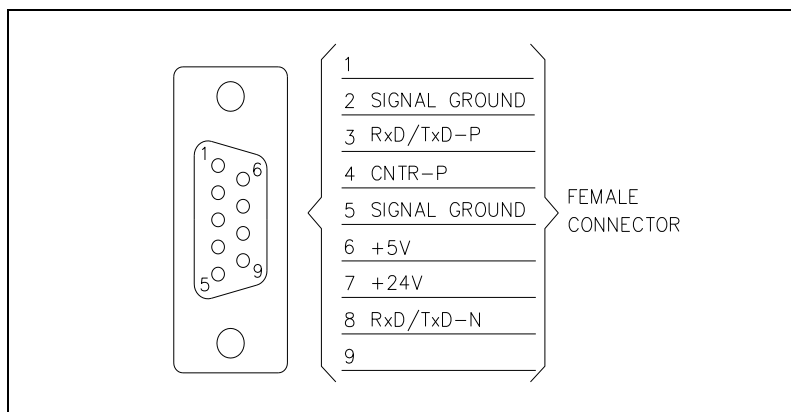


Figure 43 Profibus connector

Configuration

Before power-on the controller set the ProfiBus address by the 2 rotary switches.

Up to 126 different addresses (from 0 to 125) can be selected. The address value sets by the 2 switches is expressed in hexadecimal notation, so value from 00 to 7D are permitted; this setting is read by the interface during the power-on phase, so any change of the switch position after the power-on is ignored.

Software Operation and State Machine

From the software point of view, the interface can be seen as two devices (one for ProfiBus and the other for the internal link) linked together.

Each device has its own state machine to manage the device operation.

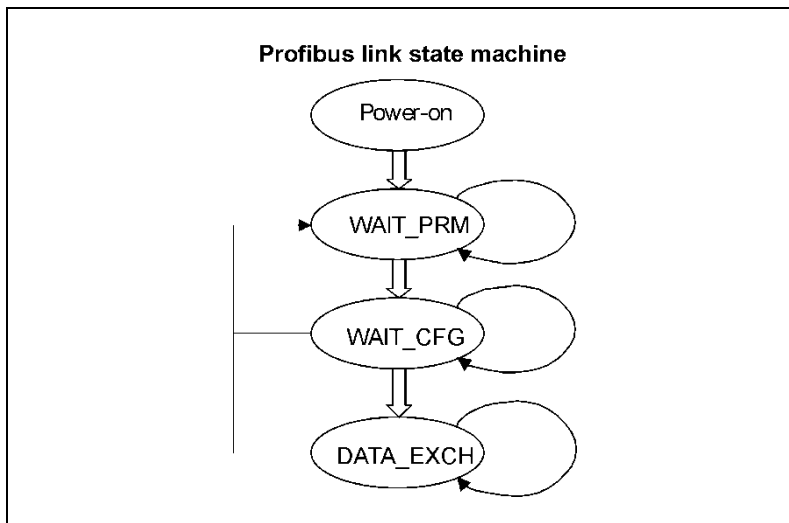


Figure 44

Tab. 17 Profibus state machine

Status	Meaning
POWER_ON	Interface set-up
WAIT_PRM	Wait for parameterisation from the master
WAIT_CFG	Wait for configuration from the master
DATA_EX	Regular data exchange

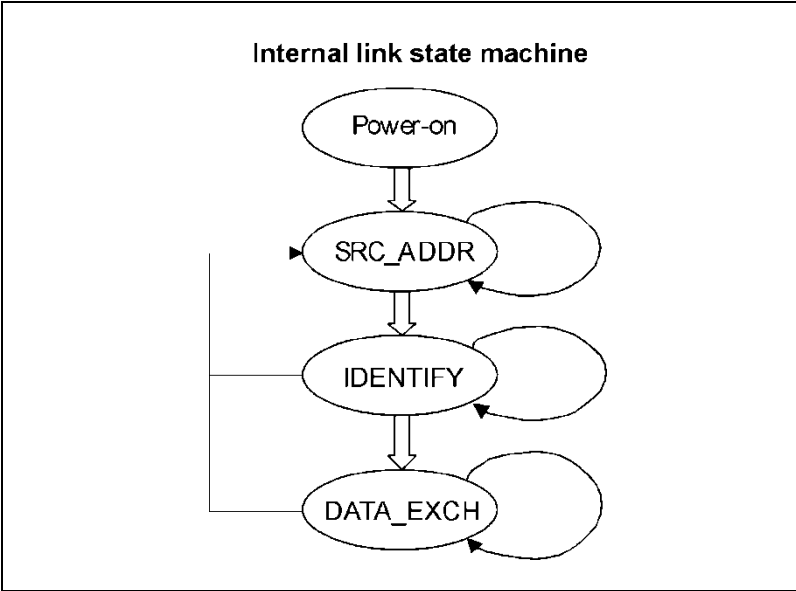


Figure 45

Tab. 18 Internal link state machine

Status	Meaning
POWER_ON	Interface set-up
SRC_ADDR	Search for the controller serial address
IDENTIFY	Identify controller model and protocol
DATA_EX	Regular data exchange

The two machines continuously share I/O data with the relative bus, and interact each other only in following conditions:

- ProfiBus Data Exchange is suspended (static diagnosis activated) if the internal serial link in not established

- Internal serial link Data Exchange is suspended (only for output data) if the Profibus State machine is not in Data Exchange mode

As a general rule, unrecoverable communication errors make each machine to jump to the state immediately following the power-on state (WAIT_PRM or SRC_ADDR).

Message Mapping

The communication is based on the continuous data exchange capability of the Profibus standard, where the master sends continuously the entire slave configuration and reads back the entire slave status.

For this purpose one Input (from slave to master) and one Output (from master to slave) buffers are provided. All the needed parameters are mapped inside each buffer.

The following tables summarise the Profibus parameter access, related to the corresponding windows serial protocol (for this protocol please refers to the serial interface paragraph).

The interface doesn't any scaling operation on the read/written parameters. It acts only as a gateway between the Profibus network and the pump's controller serial interface. As a consequence of this, the scale and the meaning of some parameters depends directly to the specific model of Turbo Pump controller connected to the interface.

Parameter Buffer

Tab. 19 Parameter buffer

OFFSET BYTE	SIZE BYTE	UNIT RES	RANGE	DESCRIPTION	WIN SERIAL
0	1	-	0-99	Gauge gas load type. The value is divided by 10 (entering 12 the controller will apply 1.2) Enter here the correction factor for gas (necessary only if gauge is connected).	161
1	2	mA/Hz/sec	0-10000	Setpoint 1 value. Enter here the threshold value for the R1 output, valid only if B ₀ -B ₂ of Parameter Buffer Offset-21 is equal to 0,1,2 or 3.	102
3	2	Sec	0-30000	Setpoint 1 Mask Time. Set here the time, measured from the start pump event, while the R1 output is masked (not activated)	103
5	2	mA/Hz/sec	0-10000	Setpoint 2 value. Enter here the threshold value for the R2 output, valid only if B ₃ -B ₅ of Parameter Buffer Offset-21 is equal to 0,1,2 or 3.	172
7	2	Sec	0-30000	Setpoint 2 Mask Time. Set here the time, measured from the start pump event, while the R2 output is masked (not activated)	173
9	2	0.2sec/bit	0-30000	Vent Valve Opening Delay. Set the delay time between the stop command and vent opening. Valid only in "purge/vent auto mode" (Parameter Buffer, Offset 20, B ₀ = 0). 0 = infinite (the valve will never open).	126
11	2	0.2sec/bit	0-30000	Vent Valve Open Time. Set how long the vent will remain open, then both purge and vent will be closed. Valid only in "purge/vent auto mode" (Parameter Buffer, Offset 20, B ₀ = 0). 0 = infinite (the valve remain open till next start command).	147
13	1	%	1-99	Setpoint 1 Hysteresis. Set the Setpoint1 hysteresis around the nominal value	105
14	1	%	1-99	Setpoint 2 Hysteresis. Set the Setpoint2 hysteresis around the nominal value	175

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Profibus Option

15	1	mBar/Torr/P a	0-99	Setpoint 1 Pressure Value Mantissa. Set the mantissa of the exponential value of the R1 setpoint (format X.XE-XX) when it is used for pressure monitoring (Parameter Buffer, Offset 21, B ₀₋₂ = 4). The value set here is divided by 10, that means 11 = 1.1	162
16	1		-10 - +10	Setpoint 1 Pressure Value Exponent. Set the exponent of the exponential value of the R1 setpoint (format X.XE-XX) when it is used for pressure monitoring (Parameter Buffer, Offset 21, B ₀₋₂ = 4).	162
17	1	mBar/Torr/P a	0-99	Setpoint 2 Pressure Value Mantissa. Set the mantissa of the exponential value of the R2 setpoint (format X.XE-XX) when it is used for pressure monitoring (Parameter Buffer, Offset 21, B ₃₋₅ = 4). The value set here is divided by 10, that means 11 = 1.1	176
18	1		-10 - +10	Setpoint 2 Pressure Value Exponent. Set the exponent of the exponential value of the R2 setpoint (format X.XE-XX) when it is used for pressure monitoring (Parameter Buffer, Offset 21, B ₃₋₅ = 4).	176
19	1			B ₀ = Stop Speed Reading. This function measures the pump speed after a stop command. 0 = Off 1 = On	167
				B ₁ = Active Stop. This function brakes the pump after a stop command, by using the motor as a generator. 0 = Off 1 = On	107
				B ₂ - Reserved	
				B ₃₋₄ = Start/Stop/Interlock Mode. Change the logic of Start/Stop and Interlock inputs. Ask to Agilent personnel for this option	151

20	1	B ₀ = Vent & Purge Auto/Man 0 = Auto (the controller drives purge and vent valves automatically) 1 = Manual (the user can command the valves by serial line or profibus)	125
		B ₁ = Gas Load Type. Select here the gas load 0 = Ar 1 = N ₂	157
		B ₂ = Interlock Type 0 = continuous (the interlock must be always present to run the pump) 1 = at start (the interlock must be present only at start command)	110
		B ₃ = Setpoint R1 Logic 0 = active close (NO) 1 = active open (NC)	104
		B ₄ = Setpoint R2 Logic 0 = active close (NO) 1 = active open (NC)	174
		B ₅₋₆ = Pressure Unit 0 = mbar 1 = Pascal 2 = Torr	163
		B ₇ = Reserved	
21	1	B ₀₋₂ = Setpoint R1 Type 0 = frequency 1 = current 2 = time 3 = normal operation 4 = pressure	101
		B ₃₋₅ = Setpoint R2 Type 0 = frequency 1 = current 2 = time 3 = normal operation 4 = pressure	171
		B ₆₋₇ = Analog Output 1 Type 0 = frequency 1 = power 2 = temperature 3 = pressure	111

Output Buffer

Tab. 20 Output buffer

Offset Byte	Size Byte	Unit Res	Range	Description	WIN Serial
0	1			B ₇ = Start/Stop 0 = Stop, 1 = Start	000
				B ₆ = Low Speed This bit set the pump target speed to "low speed value" (Output Buffer, Offset 3), otherwise the target speed is set to "high speed value" (Output Buffer, Offset 1). 0 = Off, 1 = On	001
				B ₅ – Soft Start This bit activates the soft start function which run-up the pump smoothly. Please activate this function only if the pump has been stopped for at least 3 months. 0 = On, 1 = Off	100
				B ₄ = Vent Valve Command 0 = Close, 1 = Open Valid only if in Par Buff, Offset-20, B ₀ =1	122
				B ₃ = Purge Valve Command 0 = Close, 1 = Open Valid only if in Par Buff, Offset-20, B ₀ =1	145
				B ₂ – Not used	
				B _{0,1} = Operating Mode These bits define from which source the controller has to get commands (readings are always available). In order to operate the controller by Profibus these bits must be set to 3. 0 = Remote I/O 1 = Serial 2 = Front panel 3 = Profibus	008

Input Buffer

Tab. 21 Input buffer

OFFSET BYTE	SIZE BYTE	UNIT RES	RANGE	DESCRIPTION	WIN SERIAL
0	1	V	0-100	Voltage Voltage provided to the motor	201
1	1	°C	0-100	Pump Bearing Temperature Read the pump upper bearing temperature If > 60 °C the controller goes into fail status and the pump is stopped	204
2	1	°C	0-100	Controller Power Supply Temperature Read the controller power section temperature if > 75 °C the controller goes in fail status "controller over-temperature"	216
3	1	°C	0-100	Controller CPU Section Temperature Read the controller CPU section temperature if > 80 °C the controller goes in fail "controller overtemperature"	211
4	1	-	0-6	B ₀₋₃ – Status 0 = Stop 1= Waiting for interlock (interlock connections on remote I/O connector missing) 2 = Ramp (the pump is increasing the speed) 3 = Auto tuning (the speed has been reduced because the gas load is higher than the Power Limit Applied) 4 = Braking (the pump is reducing the speed using the motor) 5 = Normal operation (the pump is at target speed) 6 = Fail (see error code byte Input Buffer Byte-27)	205

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Profibus Option

5	1	-	-	Error Code (bit field)	206
				B ₇ – Too High Load (the gas load is too high for the pump)	
				B ₆ – Short Circuit (a short circuit happened between the two of the three motor phases)	
				B ₅ – Over Voltage (too high bus voltage during braking)	
				B ₄ – Verify Line Select Not used	
				B ₃ – Power Fail (internal circuitry failure)	
				B ₂ – Controller Over-Temperature (“Controller Power Section Temperature” or “Controller CPU Section Temperature” are over the limit)	
				B ₁ - Pump Over-Temperature (“Pump Bearing Temperature” or “Pump Body Temperature” are over the limit)	
				B ₀ – Check Connection To Pump (the connection between controller and pump missing)	
6	1	-	-	B ₀ – Set Point R1 Status 0 = open 1 = closed	221
				B ₁ – Set Point R2 Status 0 = open 1 = closed	225
				B ₂ – Vent Valve Status 0 = closed 1 = open	122
				B ₄ – Not Used	
				B ₅ – Not Used	
				B ₆ – Not Used	
				B ₇ – Not Used	

7	2	mA	0-10000	Current Motor current consumption	200
9	2	W	0-200	Power Motor power adsorption	202
11	2	W	0-200	Power Limit Applied Maximum allowable power	155
13	2	Hz	0-1500	Driving Frequency Read the rotational speed of the pump	203
15	2	Hz	0-1500	Target Frequency Read the target rotational speed	120
17	2	-	0-65536	Cycle Number Number of cycles (start and stop) done by the pump	301
19	2	min	0-65536	Last Cycle Time How long the last cycle lasted	300
21	2	h	0-65536	Pump Life Total pump running time	302
23	4	mBar/Torr/ Pa	-	Pressure Reading Reports the pressure reading in the "pressure unit" selected by Par. Buff., Offset-20, B _{5.6}	224

Diagnostic Management

In addition to the ProfiBus six byte standard diagnostic, the interface provides both user diagnostic and static diagnostics functions

The standard diagnostic management is fully compliant with the ProfiBus specification.

During the start-up phase after power on, the first byte in the user diagnostic area, can assume values different from zero.

This byte has the following meaning:

- Bit 0 = 1: controller not responds (RS-485 link broken)
- Bit 1 = 1: controller not identified (not compatible with the ProfiBus gateway)
- Bit 2 = 1: first loop of requests not ended (values in the input data are not fully coherent yet)

So, if the first byte is equal to 0, the interface is working properly, otherwise there is a problem.

Each nibble in the bytes following the interface status byte represents the status of the last attempt to write a parameter with following coding (diag status byte):

- | | |
|---|---|
| 0 | Write success |
| 1 | Controller response was "NACK" |
| 2 | Controller response was "Unknown window" |
| 3 | Controller response was "Bad Data Type" |
| 4 | Controller response was "Over range" |
| 5 | Controller response was "Bad operation" |
| 6 | Controller response was an unknown response |
| F | Parameter not supported by the controller |

The ProfiBus external diagnosis services are used to report following unexpected situations:

- **Controller fail:** if the controller goes in fail mode, an external diagnostic service is required and one byte in the diagnostic buffer signals the type of failure.
- **User parameter mismatch:** if the user sets one or more parameters out of their allowed range, a ProfiBus external diagnostic service is required and some bits in the diagnosis frame signals which parameter is wrong (i.e. over range or under range).

Diagnostic Buffer

Tab. 22 Diagnostic buffer

OFF SET	MEANING	Standard diagnostic
0	B ₀ station non exist (set by master)	
	B ₁ station not ready (slave not ready for data exchange)	
	B ₂ cfg fault (configuration data doesn't match)	
	B ₃ ext diag (slave has external diagnostic data)	
	B ₄ not supported (slave doesn't support requested function)	
	B ₅ invalid slave response (slave sets permanent '0')	
	B ₆ prm fault (wrong parameter assignment)	
	B ₇ master_lock (slave is parameterised by another master)	
1	B ₀ prm req (slave has to be re-parameterised)	
	B ₁ stat diag (static diagnosis)	
	B ₂ fixed '1'	
	B ₃ wd_on (response monitoring active)	
	B ₄ Freeze mode (received freeze command)	
	B ₅ Synch mode (received synch command)	
	B ₆ reserved	
	B ₇ diag deactivated (slave is parameterised by another master)	
2	Reserved	
3	Master Add (master address after parameterisation. FF= without parameter)	
4	Ident number high	
5	Ident number low	
6	External diagnosis "Header" length indication including header	

OFF SET	MEANING	
7	Interface Status	External diagnostic
	B ₀ Controller not responds	
	B ₁ Controller not identified	
	B ₂ Input data not fully coherent yet	
	B ₃ Not used	
	B ₄₋₇ Not used	
8	B _{7.4} – Start/Stop diag status	
	B _{3.0} – Low Speed diag status	
9	B _{7.4} – Soft Start diag status	
	B _{3.0} – Vent Valve	
10	B _{7.4} – Not used	
	B _{3.0} – Mode	
11	B _{7.4} – High speed	
	B _{3.0} – Low Speed	

Error Messages

For a certain type of failure, the controller will self-diagnose the error and the following messages will be displayed.

NOTE

If the pump is not connected, the display will be as shown in the following figure.

C	H	E	C	K		C	O	N	N	E	C	T	I	O	N
			T	O		P	U	M	P						

Check connection between controller and pump, then press STOP push-button twice to start the pump.

NOTE

If the P1 input connector is not in position with the link or the external interlock connections are open, when the START pushbutton is pressed the display will be as shown in the following figure.

					W	A	I	T	I	N	G				
				I	N	T	E	R	L	O	C	K			

Disconnect connector P1 and check the link or the external interlock, then install the connector to start the pump.

NOTE

If within each step of the soft start mode the rotational speed of the pump do not reach the planned value within 15 minutes the display will be as shown in the following figure.

	T	O	O		H	I	G	H		L	O	A	D		

Verify that systems has no leaks than press STOP RESET pushbutton twice to start the pump.

Check that pump rotor is free to rotate then press the STOP RESET pushbutton twice to start the pump.

NOTE

If in normal operation (after the starting phase) the current drawn by the pump is higher than programmed (1.8 A), the pump and the interconnected devices are switched off and the display will be as shown in the following figure.

	T	O	O		H	I	G	H		L	O	A	D		

Verify that systems has no leaks than press STOP RESET pushbutton twice to start the pump.

Check that pump rotor is free to rotate then press the STOP RESET pushbutton twice to start the pump.

NOTE

If the upper bearing/pump temperature exceeds 60°C, the pump is shut off, and the display will be as shown in the following figure.

						P	U	M	P						
				O	V	E	R	T	E	M	P				

The message will stay on until the temperature decreases below threshold value. Press the STOP RESET pushbutton twice to start the pump.

17 Technical Information

Error Messages

NOTE

If the controller transformer temperature exceeds 90°C, the pump is shut off, and the display will be as shown in the following figure.

			C	O	N	T	R	O	L	L	E	R			
				O	V	E	R	T	E	M	P				

The message will stay on until the temperature decreases below threshold value. Press the STOP RESET pushbutton twice to start the pump.

			P	O	W	E	R		F	A	I	L			

Indicates a fault in the circuit of the controller used to power the pump.

NOTE

If the pump is stopped by an emergency stop signal provided via a remote contact, the display will be as shown in the following figure.

				S	Y	S	T	E	M						
			O	V	E	R	R	I	D	E					

Remove the controller power cable and check the emergency condition.

Then reconnect the power cable and press the START pushbutton to start the pump.

		O	V	E	R	V	O	L	T	A	G	E			

Controller power supply circuitry is faulty, or the Controller received a spike.

Press the START push-button twice to start the pump. Should the message still be present, call the Agilent service.

	S	H	O	R	T		C	I	R	C	U	I	T		

Check connections and shortages between pump and controller, then press the STOP RESET pushbutton twice to start the pump.

Orderable Parts

Tab. 23

Description	Part number
TwisTorr 74 FS AG Rack Controller with RS-232/485	X3508-64301
74 FS AG Rack Controller with Profibus option	X3506-64022
Air cooling kit (0.5m)	9699290
Air cooling kit Extension cable (5m)	9699940
Vent Valve N.O. 0,5 mm orifice (0,5 m cable)	9699844
Vent Valve extention cable (5 m)	9699941
Mains cable NEMA plug (3 m)	9699958
Mains cable european plug (3 m)	9699957
FRG 700/702	"see catalog"
IMG 500	"see catalog"
PVG 500/502	"see catalog"
IM 500	"see catalog"
CDG 500 T1000	"see catalog"
CDG 500 T0100	"see catalog"
CDG 500 T0010	"see catalog"
CDG 500 T0001	"see catalog"



Vacuum Products Division

Dear Customer,

Thank you for purchasing an Agilent vacuum product. At Agilent Vacuum Products Division we make every effort to ensure that you will be satisfied with the product and/or service you have purchased.

As part of our Continuous Improvement effort, we ask that you report to us any problem you may have had with the purchase or operation of our products. On the back side you find a Corrective Action request form that you may fill out in the first part and return to us.

This form is intended to supplement normal lines of communications and to resolve problems that existing systems are not addressing in an adequate or timely manner.

Upon receipt of your Corrective Action Request we will determine the Root Cause of the problem and take the necessary actions to eliminate it. You will be contacted by one of our employees who will review the problem with you and update you, with the second part of the same form, on our actions.

Your business is very important to us. Please, take the time and let us know how we can improve.

Sincerely,

A handwritten signature in black ink, appearing to read "Giampaolo LEVI".

Giampaolo LEVI

*Vice President and General Manager
Agilent Vacuum Products Division*

Note: Fax or mail the Customer Request for Action (see backside page) to Agilent Vacuum Products Division (Torino) – Quality Assurance or to your nearest Agilent representative for onward transmission to the same address.

CUSTOMER REQUEST FOR CORRECTIVE / PREVENTIVE / IMPROVEMENT ACTION

TO: AGILENT VACUUM PRODUCTS DIVISION TORINO – QUALITY ASSURANCE FAX

N°: XXXX-011-9979350

ADDRESS: AGILENT TECHNOLOGIES ITALIA S.p.A. – Vacuum Products Division –
Via F.lli Varian, 54 – 10040 Leini (TO) – Italy

E-MAIL: vpd-qualityassurance_pdl-ext@agilent.com

NAME	COMPANY	FUNCTION
ADDRESS:		
TEL. N° : FAX N° :		
E-MAIL:		
PROBLEM / SUGGESTION :		
REFERENCE INFORMATION (model n°, serial n°, ordering information, time to failure after installation, etc.):		
DATE		
CORRECTIVE ACTION PLAN / ACTUATION (by AGILENT VPD)		LOG N°

XXX = Code for dialing Italy from your country (es. 01139 from USA; 00139 from Japan, etc.)

Vacuum Products Division

Instructions for returning products

Dear Customer,

Please follow these instructions whenever one of our products needs to be returned.

Complete the attached **Request for Return form** and send it to Agilent Technologies (see below), taking particular care to include the completed **Health and Safety** declaration Section. No work can be started on your unit until we receive a completed copy of this form.

After evaluating the information, Agilent Technologies will provide you with a **Return Authorization (RA) number** via email or fax, as requested. Note: Depending on the type of return, a Purchase Order may be required at the time **the Request for Return is submitted**. We will quote any necessary services (evaluation, repair, special cleaning, eg).

Product preparation

- o Remove all accessories from the core product (e.g. inlet screens, vent valves).
- o Prior to shipment and if applicable for your product, drain any oils or other liquids, purge or flush all gasses, and wipe off any excess residue.
- o If ordering an Advance Exchange product, please use the packaging from the Advance Exchange to return the defective product.
- o Seal the product in a plastic bag, and package product carefully to avoid damage in transit. You are responsible for loss or damage in transit.
- o Include a copy of the Health and Safety Declaration in the shipping documentation on the outside of the shipping box of your returning product.
- o Clearly label package with RA number. Using the shipping label provided will ensure the proper address and RA number are on the package. Packages shipped to Agilent without a RA clearly written on the outside cannot be accepted and will be returned.
- o Return only products for which the RA was issued.

Shipping

- o Ship to the location specified on the printable label, which will be sent, along with the RA number, as soon as we have received all of the required information. Customer is responsible for freight charges on returning product.
- o Return shipments must comply with all applicable Shipping Regulations (IATA, DOT, ADR, etc.) and carrier requirements.

RETURN THE COMPLETED REQUEST FOR RETURN FORM TO YOUR NEAREST LOCATION:

EUROPE:

Fax: 00 39 011 9979 330
Fax Free: 00 800 345 345 00
Toll Free: 00 800 234 234 00

NORTH AMERICA:

Fax: 1 781 860 9252
Toll Free: 800 882 7426, Option 3

PACIFIC RIM:

Please visit our website for individual office information

vpt-customer@agilent.com

vpl-ra@agilent.com

<http://www.agilent.com>

TERMS AND CONDITIONS

Please read the terms and conditions below as they apply to all returns and are in addition to the Agilent Technologies Vacuum Product Division – Products and Services Terms of Sale.

- o Unless otherwise pre-negotiated, customer is responsible for the freight charges for the returning product. Return shipments must comply with all applicable **Shipping Regulations** (IATA, DOT, etc.) and carrier requirements.
- o Agilent Technologies is not responsible for returning customer provided packaging or containers.
- o Customers receiving an Advance Exchange product agree to return the defective, rebuildable part to Agilent Technologies **within 15 business days**. Failure to do so, or returning a non-rebuildable part (crashed), will result in an invoice for the non-returned/non-rebuildable part.
- o Returns for credit toward the purchase of new or refurbished Products are subject to prior Agilent approval and may incur a restocking fee. Please reference the original purchase order number.
- o Units returned for evaluation will be evaluated, and a quote for repair will be issued. If you choose to have the unit repaired, the cost of the evaluation will be deducted from the final repair pricing. A Purchase Order for the final repair price should be issued within 3 weeks of quotation date. Units without a Purchase Order for repair will be returned to the customer, and the evaluation fee will be invoiced.
- o Products returned that have not been drained from oil will be disposed.
- o A Special Cleaning fee will apply to all exposed products.
- o If requesting a calibration service, units must be functionally capable of being calibrated.



Customer information		
Company :		Contact Name:
Address:		Tel: Fax:
		Email:

Equipment			
Product description	Agilent PartNo	Agilent Serial No	Original Purchasing Reference
Failure description		Type of process (for which the equipment was used)	

Type of return	
☐ Non Billable ☐ Billable  New PO # (hard copy must be submitted with this form): _____	
☐ Exchange ☐ Repair ☐ Upgrade ☐ Demo ☐ Calibration ☐ Evaluation ☐ Return for Credit	

Health and safety		Substances (please refer to MSDS forms)			
The product has been exposed to the following substances: (by selecting 'YES' you MUST complete the table to the right)		* Agilent will not accept delivery of any product that is exposed to radioactive, biological, explosive substances or dioxins, PCB's without written evidence of decontamination.			
		Trade name	Chemical name	Chemical Symbol	CAS Number
Toxic	☐ YES ☐ NO				
Harmful	☐ YES ☐ NO				
Corrosive	☐ YES ☐ NO				
Reactive	☐ YES ☐ NO				
Flammable	☐ YES ☐ NO				
Explosive (*)	☐ YES ☐ NO				
Radioactive (*)	☐ YES ☐ NO				
Biological (*)	☐ YES ☐ NO				
Oxidizing	☐ YES ☐ NO				
Sensitizer	☐ YES ☐ NO				
Other dangerous substances	☐ YES ☐ NO				

Goods preparation	
If you have replied YES to one of the above questions. Has the product been purged? ☐ YES ☐ NO	
If yes, which cleaning agent/method:	
Has the product been drained from oil? ☐ YES ☐ NOT APPLICABLE	
I confirm to place this declaration on the outside of the shipping box. ☐	

I declare that the above information is true and complete to the best of my knowledge and belief. I understand and agree to the terms and conditions on page 2 of this document.	
Name:	Authorized Signature:
Position:	
Date:	
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